

CHAPTER 8

Radio Echo Theory

The theory of radio echoes from meteor trails is ancillary to the basic physical theory of meteors and arises from the need to interpret the special properties of radio observing methods. We shall use the same elementary model that was introduced in Chap. 7 with the same fundamental physical assumptions about the meteor trail and its environment. It follows that the present extension of the discussion might appropriately be called the classical radio echo theory. This chapter deals mainly with backscatter-echo phenomena; Chap. 9 will consider the extension to the forward-scatter case.

Electromagnetic radiation is involved in both the optical and the radio methods. The difference is that in the optical case the radiation is incoherent since it comes from a myriad of independent oscillators, namely, the excited and ionized atoms of the trail. The received wave is the scalar or power sum of the outputs from the individual oscillators, which have not only random phases but also different frequencies. In the radio method, one coherent wavefront from a distant oscillator illuminates all parts of the trail in turn. The wavelet scattered from any given electron is in phase with the incident wave passing that particular point of the trail. The total reflected wave is the vector or field-strength sum of many such coherent and monochromatic wavelets. The resultant aspect sensitivity of the trail, which can be both an asset and a limitation, is but one of several new facets of meteor research which have been introduced by the radio techniques. This particular property happens to be quite useful in determining the geometrical path of the meteor, but it makes it harder to trace out the complete ionization curve in the way the light curve can be recorded by the camera.

Our discussion here of the classical radio echo theory will be couched in fairly elementary terms. In a general text it is inadvisable to reproduce in any detail the rather difficult mathematical arguments dealing with certain aspects of the subject, particularly when a redrafting of the theory may be required in the light of some recent observational evidence. Several competent applied mathematicians have taken an active interest

in these analytical developments, and on occasion the aid of powerful electronic computers has been invoked to solve complicated or intractable equations. Perhaps it would be both premature and unjust to suggest that some of this advanced work may have to be discarded—premature because we are not yet sure what impact the concept of the fragmenting meteoroid may have on the classical radio theory (for that matter we are not yet positive that the faint radio meteors actually are dustballs) and unjust because no first-class creative efforts are ever really wasted and only by examining the properties of the model in detail, painstaking and laborious as the process may be, can we discover its shortcomings and devise a better approximation. In recapitulating the findings of those who have mined this field before, we shall therefore restrain the temptation to stray down some of the more rugged mathematical passages in search of higher-grade ore, not so much because of the thought that these shafts would be the first to be sealed off if the direction of the main vein were to change, but because an adequate exploration would demand more space than we can afford in this book. It must suffice to examine the nuggets uncovered in these less accessible tunnels without partaking in the arduous operations which led to their discovery.

8-1. The basic model of the underdense trail. The first step in developing the simple model is to assume that in the wake of the meteor a stationary column of free electrons is created, with a diameter that is small in comparison with the wavelength. The column does not expand radially, nor do the electrons recombine, attach, or diffuse. The trail, once created, extends from $s = -\infty$ to $s = +\infty$, where s is distance measured along the meteor path relative to an origin at the minimum range point (R_0, t_0) [see Fig. 8-1; we have $R^2 = R_0^2 + s^2 = R_0^2 + V^2(t - t_0)^2$]. The incident radio wave penetrates the column and is scattered by the individual free electrons, which oscillate freely in the applied field without colliding with other particles to any great extent. Each electron behaves as if no other were present—secondary radiative and absorptive effects may be neglected. This condition defines the *underdense* trail. The equivalent echoing area, or scattering cross section of a free electron is $\sigma_e = 4\pi e^4/m^2c^4 \sin^2 \gamma = 4\pi r_e^2 \sin^2 \gamma$, where e , m , and r_e are the charge, mass, and classical radius of the electron, c is the velocity of light, and γ is the angle between the electric vector of the incident wave and the line of sight to the receiver. For backscatter, $\gamma = 90^\circ$ and $\sigma_e \simeq 1 \times 10^{-28} \text{ m}^2$. [The constant in the older classical textbook expression for the electron cross section is $8\pi/3$, but this should be multiplied by 1.5 because the electron backscatters like a Hertzian dipole (see Sec. 2-9).]

At a point on the trail a distance R from the radio transmitter, the power flux of the incident wave is $P_T G/4\pi R^2$ watts/m², where P_T is the transmitted power and G is the antenna gain in that direction, relative

to an isotropic radiator. Assuming that the same or an identical antenna is used for reception, the effective absorbing area of the antenna is $G\lambda^2/4\pi$ when the antenna impedance is matched to the receiver. Thus the power delivered to the receiver by the wavelets scattered from a single trail electron is

$$\Delta P_R = \frac{P_T G}{4\pi R^2} \frac{\sigma_e}{4\pi R^2} \frac{G\lambda^2}{4\pi} = \frac{P_T G^2 \lambda^2 \sigma_e}{64\pi^3 R^4} \quad (8-1)$$

Since all the electrons in a line element ds of the trail will scatter in phase, we must add their field vectors rather than their power fluxes. At the receiver input the *peak* amplitude of the field vector due to a single

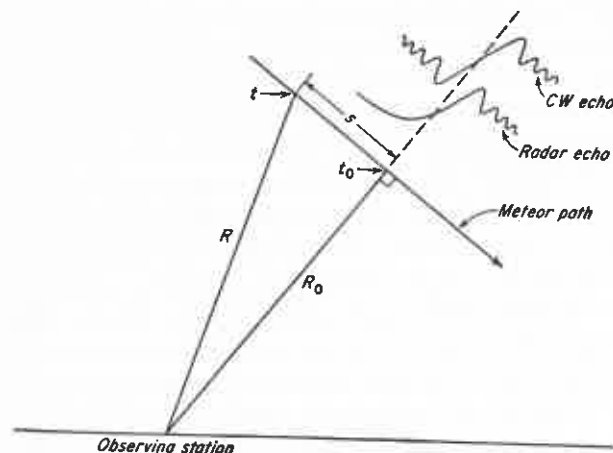


FIG. 8-1. The geometry of the meteor path relative to the observing station.

scattering electron in the trail is $(2r \Delta P_R)^{1/2}$, where r is the receiver input impedance. The absolute phase of the received wave, which has travelled a distance $2R$, is included in the time-varying expression $\sin(2\pi ft - 4\pi R/\lambda)$. As R changes with time, a phase modulation of the wave is produced (Sec. 2-7). Hence we may write

$$dA_R = (2r \Delta P_R)^{1/2} q \sin\left(2\pi ft - \frac{4\pi R}{\lambda}\right) ds \quad (8-2)$$

as the instantaneous amplitude of the echo signal received from all the electrons in the element ds , where q is the number per meter of path. The total field due to all the electrons in the trail between s_1 and s is then

$$A_R = (2r \Delta P_R)^{1/2} q \int_{s_1}^s \sin\left(2\pi ft - \frac{4\pi R}{\lambda}\right) ds \quad (8-3)$$

The parameter q may be taken out from the integral sign only if we assume it to be constant along the trail.

This integral is difficult to evaluate in general. The range changes very slowly with time near the t_0 point, though; so we can make the integral more tractable in this region by using the approximation $R \simeq R_0 + s^2/2R_0$. For convenience, we employ the transformations $\chi = 2\pi ft - 4\pi R_0/\lambda$ and $2s = x(R_0\lambda)^{1/2}$ to obtain

$$A_R = \frac{(2r \Delta P_R R_0 \lambda)^{1/2}}{2} q \int_{x_1}^x \sin\left(\chi - \frac{\pi x^2}{2}\right) dx \quad (8-4)$$

$$\text{or} \quad A_R = \frac{(2r \Delta P_R R_0 \lambda)^{1/2}}{2} q (C \sin \chi - S \cos \chi) \quad (8-5)$$

$$\text{where} \quad C = \int_{x_1}^x \cos \frac{\pi x^2}{2} dx \quad \text{and} \quad S = \int_{x_1}^x \sin \frac{\pi x^2}{2} dx$$

are the conventional Fresnel integrals of optical diffraction theory.

The maximum frequency with which the functions C and S vary with time is much less than the radio frequency f . We may therefore take a time average over an interval that is short compared with the smallest fluctuation period of C or S to obtain an expression for the quasi-instantaneous power P_R received from all the trail electrons between s_1 and s :

$$P_R = \frac{A_R^2}{2r} = \frac{\Delta P_R R_0 \lambda}{2} \left[\frac{C^2 + S^2}{2} \right] q^2 \quad (8-6)$$

or, on substituting for ΔP_R from Eq. (8-1),

$$\begin{aligned} P_R &= \frac{P_T G^2 \lambda^2 \sigma_e}{128\pi^3 R_0^3} \left[\frac{C^2 + S^2}{2} \right] q^2 \\ &= 2.5 \times 10^{-32} P_T G^2 \left(\frac{\lambda}{R_0} \right)^3 \left[\frac{C^2 + S^2}{2} \right] q^2 \quad \text{watts} \end{aligned} \quad (8-7)$$

This is the basic expression for the echo power received from the model underdense trail during the formative stage.

The bracket in Eq. (8-6) has the value unity when evaluated from $s = -\infty$ to $+\infty$. The approximation that we used for R does not permit us to go very far in either direction from the t_0 point, of course, but it may be shown from tables of C and S (Jahnke and Emde, 1938) that the contributions from the remoter parts of the trail are negligible and that the value of the bracket is close to unity when taken over a few Fresnel zones on both sides of t_0 .

The form of Eq. (8-6), with the bracket equal to unity, was first derived by Blackett and Lovell (1941) for the power scattered by a line of electrons, although at the time they were anticipating the possibility that the ionized line might be created by a cosmic-ray particle. Later work by

Lovell (1947, 1950) and by Lovell and Clegg (1948) extended the theory to the meteor case.

8-2. The diffraction echo. If we assume all the parameters in Eq. (8-7) to be constant for a given meteor we may write a simplified form of the equation as

$$P_r = F_r^2(C^2 + S^2) \quad (8-8)$$

where $2F_r^2$ is the final echo-power level. The subscript r now refers to the power received from the meteor-reflected wave. This expression is identical with that which describes the optical pattern of diffraction fringes produced at a straight edge.

The detector may also receive a small fraction of the transmitted power directly. Mr. E. L. R. Webb of the NRC developed the analysis of this case, which applies to the simple cw system wherein transmitter and receiver are spaced a short distance D , just great enough to keep the receiver from being overloaded with the direct ground wave. We then have an additional field vector present at the receiver, defined by $B_d \sin [2\pi ft - (2\pi D/\lambda)]$, which must be added to the vector A_r in Eq. (8-5) before taking the short-term time average. When this is done, the total power at any moment is given by

$$P_{r+d} = F_r^2(C^2 + S^2) + F_d^2 + 2F_r F_d (C \cos \psi - S \sin \psi) \quad (8-9)$$

$$\text{or} \quad P_{r+d} = (F_r C + F_d \cos \psi)^2 + (F_r S - F_d \sin \psi)^2 \quad (8-10)$$

where F_d^2 is the power level of the direct wave and $\psi = 2\pi(2R_0 - D)/\lambda$ is the phase difference between the direct and reflected paths at time t_0 .

The expression for P_r given by Eq. (8-8) increases slowly as t_0 is approached. As the meteor crosses t_0 the power rises quickly to a maximum; after this, oscillations of decreasing amplitude and increasing frequency occur about the final steady-state level. This is the ordinary noncoherent radar case. It appears in Fig. 8-2a as the dashed curve (see also Fig. 8-9, curve A). The square root of the power, i.e., the echo amplitude, has been plotted here because the ordinary radar receiver output is usually linear with voltage rather than with power. This theoretical curve may be compared with the observed radar records (Fig. 4-19).

For the cw case, Eq. (8-9) or (8-10), oscillations of increasing amplitude and decreasing frequency appear before t_0 , as is shown by the solid curves, Fig. 8-2a to h, which have been plotted for selected values of the parameter ψ . The post- t_0 oscillations are also present. The value of ψ will determine whether the post- t_0 receiver output level is above or below that of the direct wave alone. These theoretical amplitude-time curves may be compared with the observed cw records (Figs. 4-9b, 4-10b, and 4-11).

A coherent detector, employing two outputs with quadrature injection

of the reference wave, would eliminate the first two terms of Eq. (8-9) and yield only the third, which would simplify some of the analysis.

It should be emphasized that high-speed recording methods are needed, with time resolutions of the order of milliseconds, to bring out the fine structure of the oscillations. The ordinary radar range-time display, or a slow paper tape recorder, shows only the sudden rise in echo strength to the final mean level, and the fine detail of the oscillations is usually compressed and indiscernible.

We have avoided the use of the notation of the complex variable, although it would have enabled us to express the analysis in this and the preceding section in more compact form, because we have endeavored to keep the physical picture as clear as possible. In fact, it may be still easier to visualize the results if Eqs. (8-8) and (8-9) are expressed in terms of more familiar functions, by means of Cauchy's approximations for the Fresnel integrals. If we exclude the small region near t_0 where $-1 < x < 1$, and also assume x_1 to be effectively at minus infinity, then the Cauchy approximations become

$$C = \frac{1}{\pi x} \sin \frac{\pi x^2}{2} \quad S = -\frac{1}{\pi x} \cos \frac{\pi x^2}{2}$$

for negative values of s , prior to t_0 , and

$$C = 1 + \frac{1}{\pi x} \sin \frac{\pi x^2}{2} \quad S = 1 - \frac{1}{\pi x} \cos \frac{\pi x^2}{2}$$

for positive values of s , after t_0 . Hence, for the ordinary radar case,

$$P_r(-s) = \frac{F_r^2}{\pi^2 x^2} \quad (8-11)$$

$$\text{and} \quad P_r(+s) = 2F_r^2 + \frac{F_r^2}{\pi^2 x^2} + \frac{2\sqrt{2}}{\pi x} F_r^2 \sin \left(\frac{\pi x^2}{2} - \frac{\pi}{4} \right) \quad (8-12)$$

which are together equivalent to Eq. (8-8). Note the slow, nonoscillatory increase prior to t_0 , followed by the rise to a maximum after t_0 , with superposed oscillations about the final level $2F_r^2$.

For the cw or coherent-pulse radar case, before t_0 ,

$$P_{r+d}(-s) = \frac{F_r^2}{\pi^2 x^2} + F_d^2 + \frac{2F_r F_d}{\pi x} \sin \left(\frac{\pi x^2}{2} + \psi \right) \quad (8-13)$$

which shows that the slow rise in power begins at the level of the direct wave F_d^2 and the diffraction oscillations appear before t_0 . After t_0 , we have

$$P_{r+d}(+s) = \frac{F_r^2}{\pi^2 x^2} + E^2 + \frac{2F_r}{\pi x} E \sin \left[\frac{\pi x^2}{2} - \tan^{-1} \left(\frac{F_r - F_d \sin \psi}{F_r + F_d \cos \psi} \right) \right] \quad (8-14)$$

where $E^2 = F_d^2 - 2\sqrt{2}F_dF_r \sin\left(\psi - \frac{\pi}{4}\right) + 2F_r^2$. Here the final power level is determined both by the relative values of F_d and F_r and by the phase angle ψ . Note that post- t_0 oscillations should always be present, unless $E = 0$. Since $E = 0$ only when $\sqrt{2}F_r = F_d$ and $\psi = 3\pi/4$ (see Fig. 8-2d) there should be very few cases of this nature. In practice, though, an appreciable number of records show little or no sign of post- t_0 oscillations—an observation that has not yet been explained satisfactorily.

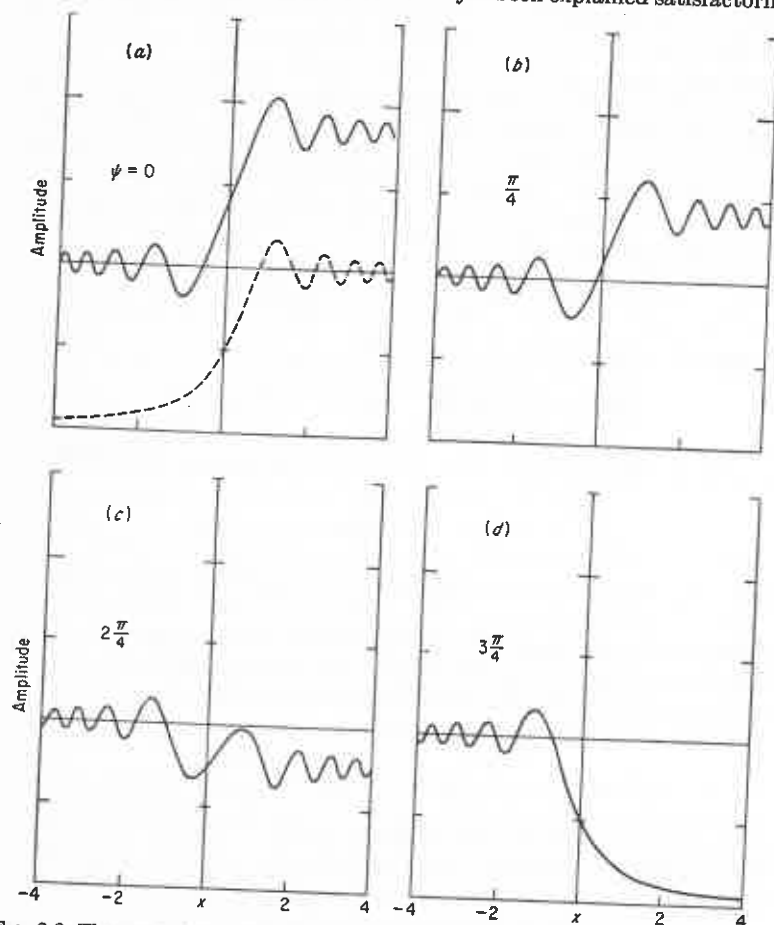
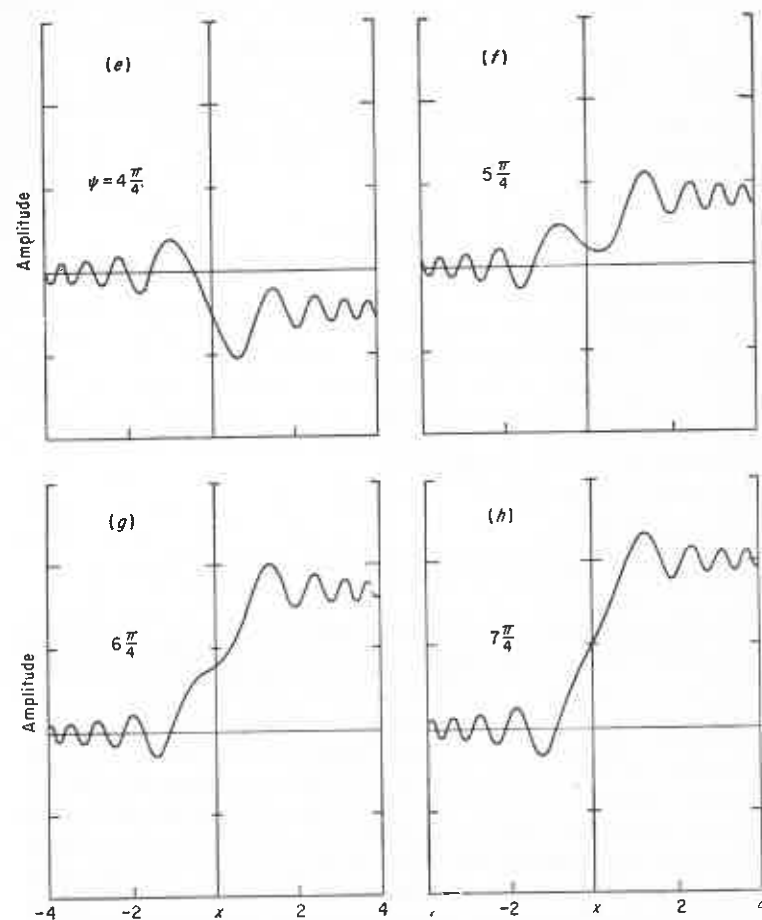


FIG. 8-2. Theoretical amplitude-time curves of meteor echoes based on the diffraction theory. Receiver output voltage is plotted against the argument x of the Fresnel integrals, which is directly proportional to time. The dashed curve in graph (a) is

A graphical representation of the diffraction echo may be obtained by plotting the C and S integrals orthogonally to produce the Cornu spiral shown in Fig. 8-3. For clarity, the variable x in the Fresnel integrals has been marked out along the curve only as far as $x = \pm 3.5$, though each spiral continues in to its center as $x \rightarrow \pm \infty$. To use the diagram, you mark the beginning and end points of the meteor trail on the curve. The line joining these points represents the total echo amplitude of that



for the noncoherent radar case. The solid curves are computed for the cw or coherent-pulse radar case, for selected values of ψ , and for equal strengths of the reference voltage and the final meteor-echo voltage.

segment of trail, and its angular orientation with respect to an arbitrary fixed direction is the echo phase. You square the length of the line to obtain the echo power.

For the ordinary noncoherent radar case the origin of the meteor echo vector is held anchored at $x_1 = -\infty$. The F_r vector grows in amplitude as its tip follows around the lower left spiral and crosses the t_0 point, while

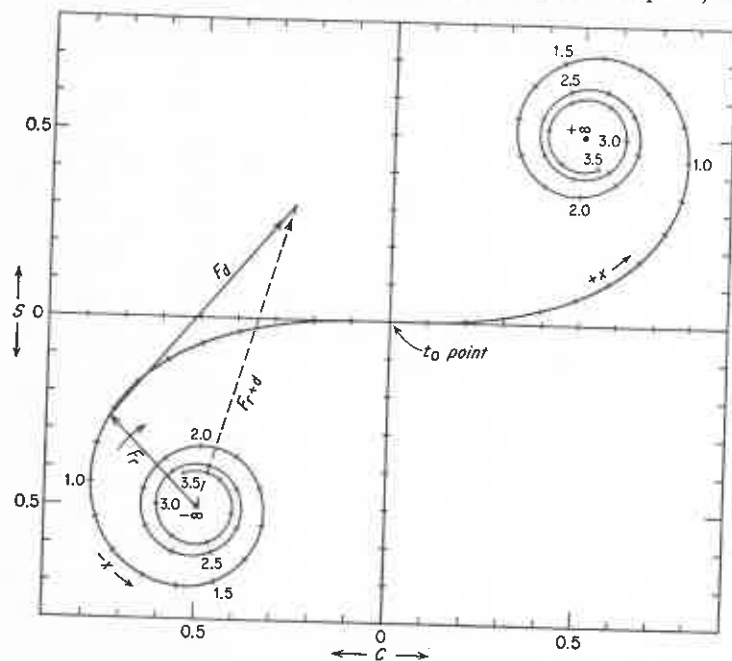


FIG. 8-3. The Cornu spiral. The meteor-echo vector F_r revolves around the point $x = -\infty$ with its tip following the spiral. When a reference wave is present, the constant vector F_d is added to F_r to yield the resultant signal amplitude F_{r+d} .

the phase angle of the vector increases continuously. The vector reaches its maximum around $x = +1.2$ as the meteor crosses the boundary of the first Fresnel zone past t_0 , and then it fluctuates in amplitude as the tip goes around the upper-right spiral, ending up at the final value given by the line joining the centers of the spirals. While the vector tip is following the upper spiral, the phase angle no longer increases continuously but oscillates back and forth over a small and decreasing angular interval. Because the phase is lost in the ordinary noncoherent detector, only the amplitude variation will be recorded. This will be identical to the dashed curve in Fig. 8-2a.

In the case of a cw or coherent-pulse radar, a constant vector F_d due to the direct wave must be added to F_r (see Fig. 8-3). The resultant F_{r+d} vector will, in general, show both amplitude and phase fluctuations prior to t_0 as well as after t_0 . An ordinary detector will reveal only the amplitude variations. The phase can be recovered if a quadrature detector is used which has two outputs, one combining F_r and F_d and the other combining F_r with F_d rotated by 90° . When the two outputs are displayed orthogonally on a cathode-ray tube, the Cornu spiral is created.

8-3. Velocity measurements. The oscillatory terms of Eqs. (8-12), (8-13), and (8-14) are the ones of chief interest in velocity measurements. In any of these terms we may replace $\pi x^2/2$ by $2\pi V^2 \tau^2 / R_0 \lambda$, where $\tau = t - t_0$. Let the instantaneous frequency f_i be defined by $\sin(2\pi f_i \tau) = \sin(2\pi V^2 \tau^2 / R_0 \lambda)$. Differentiating both sides with respect to τ we have

$$f_i = \frac{2V^2}{R_0 \lambda} \tau \quad (8-15)$$

Thus the slope of a graph of f_i against τ is proportional to the square of the meteor velocity. Manning (1948) deduced the same expression for a "moving-ball" type of target. Actually, for the purpose of measuring velocities, there is little practical difference between the moving-ball interference theory and the cw line-diffraction theory, except near t_0 . The moving-ball curves are symmetrical with respect to t_0 and show no post- t_0 shift (see Fig. 8-9, curve D), whereas the diffraction curves are in effect antisymmetrical with respect to t_0 (see Fig. 8-2) and are usually translated up or down after t_0 .

If one excludes a few cycles near t_0 , the observed relationship between f_i and τ is linear to less than one per cent for the better diffraction-echo records, some of which may show several hundred cycles prior to t_0 . However, most small meteors produce only a few oscillations near t_0 , where use of this technique can lead to rather large errors. Fortunately, the approximate position of the t_0 point can usually be located on the record, particularly in the cw method where the oscillations often appear on both sides of t_0 . The radial component of wind velocity can cause some distortion of the diffraction waveform in the vicinity of t_0 , but this effect is serious only for slow meteors and strong radial winds. The time taken by the average meteor to cross the first Fresnel zone is usually much less than the period of the body-Doppler fluctuations caused by wind drift. In most cases the phase angle ψ at t_0 can be determined to within $\pm 45^\circ$ from the record (rarely should the error be as great as $\pm 90^\circ$) by comparison of the observed waveform with a catalog of calculated waveforms covering suitable ranges of ψ and of the ratio F_r/F_d . Starting

from t_0 , one can then count cycles backward in time. Let $\Delta\tau$ be the time interval between the m th and n th cycles (measured when feasible at the zero-crossing points where the slope is steepest—otherwise at the maxima). We have $\Delta\tau = t_n - t_m = (\sqrt{nR_0\lambda} - \sqrt{mR_0\lambda})/V$, or

$$V = \sqrt{R_0\lambda} \frac{\sqrt{n} - \sqrt{m}}{\Delta\tau} \quad (8-16)$$

For convenience in reduction one may use perfect squares for m and n , e.g., the 4th and 9th cycles or the 9th and 16th cycles. The velocity error due to the uncertainty in ψ can easily be computed. The m th and n th zero crossings occur when $\pi x^2/2 \pm \psi$ equals $2\pi m$ and $2\pi n$, respectively. Let us take the particular case $m = 4$, $n = 9$ which was used to a great extent in reducing the Ottawa records. When $\psi = 0$ or $\pm 2\pi$, the expression $V \Delta\tau/\sqrt{R_0\lambda}$ equals unity and yields the correct value of V from the measured $\Delta\tau$. When ψ is near $\pm 90^\circ$, the values of $V \Delta\tau/\sqrt{R_0\lambda}$, measured at the zero-crossing points, can range from 0.98 to 1.02, corresponding to an error in V of ± 2 per cent. The correct value for V at $\psi = \pm 180^\circ$ can be found by turning the record over (thereby effectively shifting the phase 180°) and reading it as before. In between the points $\psi = 0$, π , and 2π , small errors in V will result from the use of the simple two-point formula, the errors decreasing for larger m and n . When $m = 4$, $n = 9$, the mean error over the whole range of ψ is $+1$ per cent if $\Delta\tau$ is measured at zero crossings, and -1 per cent if the maxima are used. A suggestion put forward by Mainstone (1960), that meteor velocities measured by an application of Eq. (8-16) will have a large systematic error on the low side, can be valid only if the analyst pays no attention at all to the waveform at t_0 and the information it contains regarding ψ .

The Jodrell Bank workers have developed similar formulas for use in measuring their radar amplitude-time records (Davies and Ellyett, 1949). In their experiments the diffraction-echo fluctuations were obtained only after t_0 and were independent of ψ .

The reduction of the diffraction-echo curves may be done automatically if the output of the receiver is fed directly into a recording frequency analyser of the type employed to register whistler phenomena on very low radio frequencies. Dr. L. A. Manning has kindly furnished some unpublished frequency-time records of meteor echoes obtained in this manner (see Fig. 8-4). The slopes of the echo curves near the t_0 points are proportional to the meteor velocities. In this particular experiment the transmitter and receiver were widely separated; hence the constant of proportionality depended both on the spacing of the stations and on the position and orientation of the trail (see Sec. 9-1). The geometry would be simplified and the ambiguities would be removed if the transmitter and receiver were placed close together, though fewer echoes would be seen. Note in Fig. 8-4 that some of the meteor echoes approach asymp-

SEC. 8-3]

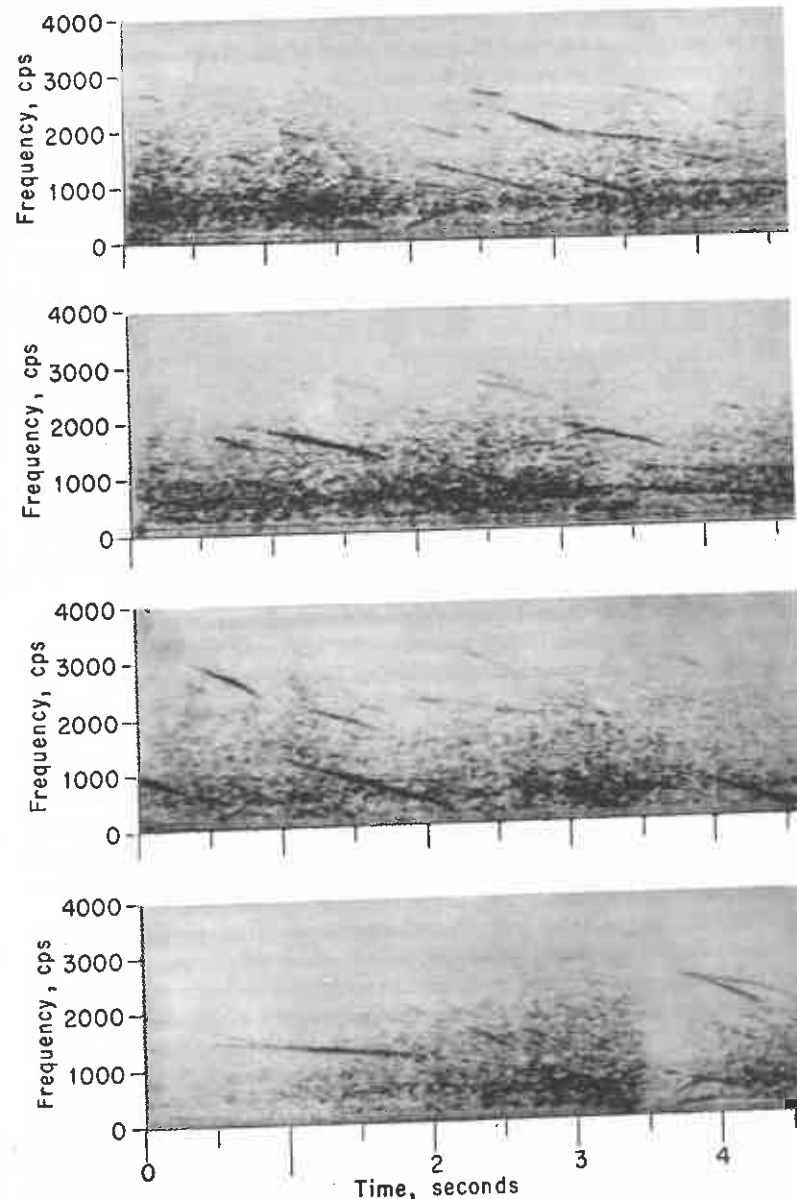


FIG. 8-4. Frequency-time records of meteor echoes obtained over a forward-scatter path. (Courtesy of L. A. Manning.)

totically to the limiting frequency corresponding to the head-on aspect of the meteor. Abnormal changes in slope of the frequency-time curve would indicate deceleration of the meteor.

This frequency-analyzing technique could be a great help in speeding up the velocity reductions, if high precision is not essential. As in the conventional records, the envelope of the amplitude-time variations can also be determined, though the frequency-time method yields this more directly. Any departure of this envelope from the theoretical envelope given by Eq. (8-8) could be interpreted in terms of variation of ionization along the path.

8-4. Modifications to the underdense trail model. The bare simplicity of our basic model requires many qualifications before it can hope to represent the complex conditions of the average meteor. We know that ionization is not distributed uniformly along the trail. The trail does expand radially, and it may have an effective initial diameter r_0 which is far from small in comparison with the radio wavelength. The effects of diffusion, recombination, attachment, turbulence, and wind shears must be considered; some of these will certainly be involved in the eventual dissipation of the trail. Let us first examine the influence these factors may have on the early history of the trail during the diffraction-echo observations.

The ions and electrons move out in an electrically neutral cloud (ambipolar diffusion). The radial expansion of the trail will proceed at a velocity of the order of the thermal velocities of the environment. This may be of the order of 0.5 km/sec, say. The meteor velocities range from 12 to 72 km/sec, and hence the trail is a narrow cone rather than a uniform cylinder, with a half-angle ranging from perhaps 2.5 to 0.5°. More accurately, the cone will not be right cylindrical but will have an exponential taper owing to the decrease in both r_0 and D with height (Sec. 8-6). This slight virtual realignment of the trail axis has a negligible second-order effect on the velocity measurement, though it would need to be taken into account in accurate radiant determinations.

Öpik (1955d) pointed out that the great penetrating power of the ablated meteor atoms, which have initial velocities comparable to that of the meteor, will very quickly create an ionized column which has an appreciable initial radius r_0 . Many collisions are required to slow the atoms down to thermal velocities, though less than a millisecond is needed to complete the process. Replacing the line of electrons by an initial column of finite width created in less than a millisecond will not affect the velocity measurement, though it does have a bearing on the echo signal strength, as we shall see later. Öpik gives a table from which the initial radius may be computed for any height in the atmosphere.

As might be expected there is a slight velocity dependence, but Öpik's mean value for stony meteors moving at 40 km/sec may be taken as $r_0 = 1.1 \times 10^{-6} \rho_0 / \rho_a$ meters, where ρ_0 is the atmosphere density at sea level. The ionized oxygen trail has been assumed here. By taking an average between the 1956 and 1959 ARDC density data in the Appendix tables we find that $\log_{10} \rho_a = 1.3 - 0.075H$ will represent the density fairly well for $75 < H < 120$ km. Hence

$$\log_{10} r_0 = 0.075H - 7.2 \quad (8-17) \quad \star$$

for the initial radius of the path, in meters. At 96 km, $r_0 = 1.0$ m.

Manning (1958) is of the opinion that the initial radius of the trail may be of the order of 14 ionic MFPs (mean free paths), again with only a slight dependence on velocity. The ionic MFP is about one-fifth of the neutral particle MFP, as listed in the Appendix tables. Since $\log_{10} \text{MFP}_{(\text{neutral})} + \log_{10} \rho_a = -7.10$ from the Appendix tables, the following relation is obtained: $\log_{10} \text{MFP}_{(\text{neutral})} = 0.075H - 8.4$. This is reasonably satisfactory for $75 < H < 120$ km. If we adopt Manning's conclusions, namely $r_0 \simeq 14 \text{MFP}_{(\text{ionic})} \simeq 3 \text{MFP}_{(\text{neutral})}$, we get

$$\log_{10} r_0 = 0.075H - 7.9 \quad (8-18)$$

If the atom trail is adopted, on the thesis that ionization can occur while the atom trail is expanding, the constant in Eq. (8-18) would be -7.2 , which agrees with that in Eq. (8-17). Experimentally, though, Eq. (8-18) as it stands seems to fit better with the evidence on ceiling heights (see Sec. 8-6).

To return to the velocity discussion let us insert some typical figures from the Ottawa experiments in Eq. (8-16). Here the mean $R_0 = 150$ km, $\lambda = 10$ m, $n = 9$, and $m = 4$. Thus, for the average meteor, $V \Delta \tau = 1.2$ km, and the $\Delta \tau$'s range from 0.017 to 0.1 sec. In this short path length, any likely variation in q could affect the echo amplitude somewhat, but it would create second-order phase errors only. Amplitude variations were indeed apparent in many of the records, particularly of the brighter meteors, though the f_i/τ curves were smooth and linear.

Wind drifts and shears can be more troublesome than any of the effects mentioned so far. Assume that a shear occurs between the path segment under observation and the t_0 point, with a radial velocity component of 10 m/sec, say. For a 72-km/sec meteor the translational shift between the $\Delta \tau$ segment and the t_0 point is $2 \times 10 \times 0.05 = 1$ m. The resultant phase shift of about 36° will add rather less than 1 per cent to the over-all velocity error. The same argument applies to the effects of a steady radial wind drift of the trail as a whole, which will create a

similar phase shift between the Δr segment and the t_0 point.* A very slow meteor, though, with five times this phase error, will lose or gain a half-cycle in the diffraction pattern between Δr and t_0 , and the velocity error may be several per cent. Kaiser (1955a) has examined the effects of wind shears in detail. In the light of our present knowledge of wind shears he concluded that the mean effect of shears on the velocity measurement of the average meteor would be about 1 per cent. Kaiser extended his analysis to show that the radiant determination errors were more seriously affected by wind shears. Average positional errors up to 3° may be ascribed to this cause.

If the contribution of the lower limit x_1 of the Fresnel integrals is not negligible, one may develop sets of predicted diffraction curves which are analogous to the optical diffraction fringes due to a rectangular slit, the slit width varying with time. Some of the observed records are clearly due to trails which have begun or ended fairly abruptly near t_0 or which have not come near t_0 . It was pointed out some time ago (McKinley, 1951b) that in a few cases rapid diffusion might cause the lower limit x_1 to move in the direction of the meteor while it is still traveling down its path. Furthermore, if x_1 were to move with about the same velocity as the meteor, the target would be effectively a short-line segment traveling across the line of sight. The effect is complicated because the exponential decrease of D with H means that an initially short segment will not remain fixed in length as the meteor descends. This gives rise to a combination of diffraction and moving-ball effects.

Considerable study has since been devoted to this problem, which becomes increasingly important as the radio wavelength is decreased. Eshleman (1953) has discussed the influence which rapid diffusion has on echo rates observed on the higher radio frequencies, and in a later, unpublished report he derived quantitative expressions for the effective echoing area of a short-segment trail. The remarks of Hawkins (1956b) on the diffusion ceiling have already been noted. Flood (1957) developed the case where the segment is much shorter than the first Fresnel zone. In this situation the Fraunhofer diffraction theory of physical optics is more appropriate than the Fresnel theory. Qualitatively, the results mean that the echo from a faint underdense meteor trail, as observed on frequencies well above 100 Mc/sec, will be considerably weaker than if the trail were assumed to be several Fresnel zones in length around t_0 . In addition, if the initial trail width is large in terms of wavelength, the

* To Mainstone (1960) goes the credit for demonstrating that the sign and magnitude of this progressive phase shift can be accurately determined by using one of the wind-drift methods described in Sec. 4-11. Furthermore, the technique allows one to remove any residual error in the determination of ψ at t_0 , due to wind drift, which can be important for slow meteors (see Sec. 8-3).

signal will be further attenuated. On the other hand, the aspect sensitivity of the trail becomes less pronounced in comparison with the long-trail case, the echo strength does not fall off as rapidly on either side of t_0 , and the echo curves become evenly symmetrical about t_0 . The next two sections will cover some of these points in more detail.

8-5. The decay of the underdense trail echo. After the meteor has finished its course and has produced a trail of effective radius r_0 , the most important factor in reducing the echo strength is ambipolar diffusion, which reduces the volume density without affecting the line density. Herlofson (1951), Huxley (1952), Kaiser (1953), and others have analyzed the problem by setting up and solving the differential equations for radial diffusion of a cylindrical distribution of electrons and positive ions, in the absence of a magnetic field.

A Gaussian distribution for the radial density of electrons will be adopted throughout this book as the most reasonable model, though other models have been treated (Herlofson, 1951; Kaiser and Closs, 1952; Eshleman, 1955). The standard form of the radial diffusion equation is

$$\frac{\partial N}{\partial t} = \frac{D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial N}{\partial r} \right) \quad (8-19)$$

where N is the volume density of electrons at time t and distance r from the axis and D is the ionic diffusion coefficient in square meters per second. One particular solution of Eq. (8-19) has the form

$$N = \frac{1}{A(t+k)} e^{-[r^2/B(t+k)]}$$

where A , B , and k are constants. Direct substitution in Eq. (8-19) shows immediately that $B = 4D$. The constant A may be found from the condition that the total number of electrons in an infinite cross-sectional slice of unit thickness must equal q at all times. The number in an annular ring of radius r is $2\pi r N dr = \pi N d(r^2)$, and hence

$$q = \int_0^\infty \frac{\pi}{A(t+k)} e^{-[r^2/4D(t+k)]} d(r^2) = \frac{4\pi D}{A}$$

Thus the volume density may be expressed as

$$N(r,t) = \frac{q}{\pi(4Dt + r_0^2)} e^{-[r^2/(4Dt + r_0^2)]} \quad (8-20)$$

where the constant $4Dk$ is replaced by r_0^2 , the square of the initial radius of the trail.

The diffusion coefficient D increases roughly exponentially with height in the meteor region. It may depend significantly on local atmospheric conditions and show daily and seasonal variations. Murray (1959)

emphasizes that the great scatter of the observational data suggests that caution should be exercised in applying the results. Perhaps the most reliable determination of D is to be found in the work of Greenhow and Neufeld (1955). Their results may be summarized, for the region between $H = 80$ km and $H = 110$ km, by the relation

$$\log_{10} D = 0.067H - 5.6 \quad (8-21)$$

where D is in square meters per second and H in kilometers. (See the Appendix for a discussion of D and other atmospheric parameters.) The decrease of both D and r_0 as the meteor descends should be taken into account in a more sophisticated treatment.

After the creation of the trail, we determine the echo amplitude by summing up the individual amplitude vectors from the electrons throughout the ionized cylinder. The time scale is now measured in tenths of seconds rather than in hundredths of seconds as in the discussion of the diffraction echo during the creative process. In the previous sections we considered that the cylinder radius was negligible in comparison with the wavelength. We now have to take account of the radial distribution of electrons, defined by Eq. (8-20), since there will be destructive interference effects between the electrons in any given cross section of the cylinder. Herlofson (1951), Eshleman (1955), and Brysk (1958), among others, have carried out this integration with varying degrees of generality. This is really a three-dimensional problem, but the essential answer can be obtained by considering the phase effects in a cross section near t_0 where the echo is strongest.

An annular ring of radial width dr is drawn about the trail axis in the cross-sectional plane (see Fig. 8-5). Throughout this ring the electron density is uniform. The amplitude of the echo signal from all electrons in this ring will be

$$dA = 2Nr dr \int_0^\pi \sin \left(2\pi ft - \frac{4\pi r}{\lambda} \cos \theta \right) d\theta$$

where the phase angle $(4\pi r/\lambda) \cos \theta$ of the electrons in the element $r dr d\theta$ is referred to a datum plane of zero phase, PP' , which is drawn normal to the line of sight of the far distant radar station. Let us expand the integrand above. Then, since

$$\int_0^\pi \cos \left(\frac{4\pi r}{\lambda} \cos \theta \right) d\theta = \pi J_0 \left(\frac{4\pi r}{\lambda} \right)$$

and the corresponding integral of the sine function is zero, we have

$$dA = 2\pi N J_0 \left(\frac{4\pi r}{\lambda} \right) r dr \sin 2\pi ft$$

SEC. 8-5]

This amplitude is less than that which would be obtained if all the electrons were concentrated at the trail axis. The resultant phase angle is always zero for any ring, though, and so we may omit the r - f term and add up the amplitude contributions from all the rings. The ratio of the total scattered amplitude from the diffusing electrons to the total amplitude which would be obtained if all the electrons were in phase at the trail axis is

$$\frac{A}{A_0} = \frac{2\pi \int_0^\infty N J_0(4\pi r/\lambda) r dr}{2\pi \int_0^\infty N r dr} \quad (8-22)$$

In Eq. (8-22) we substitute the value of N given by Eq. (8-20). The denominator, of course, is equal to q . The numerator is a little more

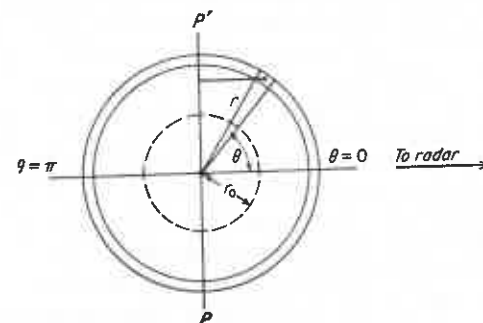


FIG. 8-5. Cross section of ionized trail near t_0 point.

difficult to evaluate* but it has been done in several texts (cf. G. N. Watson's "Theory of Bessel Functions," Cambridge University Press, New York, 1952, p. 393, or "Formulas and Theorems for the Special Functions of Mathematical Physics" by W. Magnus and F. Oberkettner, Chelsea Publishing Company, New York, 1949, p. 35). When the integration is carried out, Eq. (8-22) becomes

$$\frac{P_R(t)}{P_R(0)} = \left(\frac{A}{A_0} \right)^2 = e^{-(32\pi^2 D t / \lambda^2)} e^{-(8\pi^2 r_0^2 / \lambda^2)} \quad (8-23)$$

where $P_R(0)$ is given by Eq. (8-7), after the meteor has passed t_0 . This shows that the received power $P_R(t)$ is immediately reduced by an initial factor $\exp(-8\pi^2 r_0^2 / \lambda^2)$ because of the finite initial width of the trail.

* The do-it-yourself reader might like to try this for his own satisfaction. One of several ways is to expand the Bessel function $J_0(4\pi r/\lambda)$ in its power series and then to integrate term by term, which results in the power series for the exponential.

The echo power subsequently decays according to the time-varying diffusion factor.

The *echo-amplitude decay time* T_{un} (or decay-time constant) for the underdense trail is defined as the time taken for $P_R(t)$ to decay by a factor of e^{-2} , or the time interval during which the echo power decreases by 8.7 db. We have, from Eq. (8-23),

$$T_{un} = \frac{\lambda^2}{16\pi^2 D} \quad (8-24)$$

This means that the theoretical decay time is independent of all the equipment parameters except the wavelength. Furthermore, T_{un} is the same for all meteors at a given height, no matter what their line densities are—provided the densities do not exceed the limit for an underdense trail. The echo decay time should not be confused with the time during which the echo is visible above a fixed threshold level, such as the receiver noise level. The latter time interval is strongly dependent on both q and the other system parameters, transmitter power, antenna gain, and receiver sensitivity. For example, with $\lambda = 9$ m and $D = 3$ m²/sec, we have $T_{un} = 0.2$ sec for all underdense trail echoes, but the duration above receiver noise could range from nearly zero for the smallest detectable meteor to a second or more for a meteor with $q \simeq 10^{14}$ electrons/m, as recorded by a typical radar system (see Fig. 8-11, *dashed curves*).

8-6. The height ceiling for the underdense echo. Let us examine in more detail the power attenuation expressed by Eq. (8-23). Figure 8-6 is a model of the trail, frozen at the instant when the meteor head has traversed the first Fresnel zone past t_0 and is at the point defined by $+\tau_F = (R_0\lambda/2)^{1/2}/V$. At this place and time the effective trail radius is just the r_0 appropriate to the height. At the same moment, but at any point farther back up the trail, the effective radius will be $(r_0^2 + 4Dt)^{1/2}$, where t is measured backward from the point $+\tau_F$. In Fig. 8-6, the fixed r_0 cone is shown as the dashed line. The expanding diffusion cone is shown as the solid envelope. We now make two assumptions which are susceptible of more rigorous verification than we propose to offer in this text. First, most of the echo signal comes from the first Fresnel zone—this is fairly obvious from an examination of the Cornu spiral, Fig. 8-3, or from Eq. (8-6). Second, the signal received from the whole trail will be approximately equal to that which would be received if the trail were to have the uniform diameter shown at the t_0 point on the diagram. The latter assumption can be verified by an involved numerical integration along and across the trail at all points.

The effective radius at the t_0 point, under the above conditions, is defined by Eq. (8-20) when $t = \tau_F = (R_0\lambda/2)^{1/2}/V$. The relative echo power may be found by taking logarithms of both sides of Eq. (8-23),

after substituting this value of t . We obtain

$$\log_{10} \frac{P_R(t)}{P_R(0)} = -0.4343 \times 32\pi^2 \frac{R_0^{1/2} D}{2^{1/2} \lambda^{3/2} V} - 0.4343 \times 8\pi^2 \left(\frac{r_0}{\lambda}\right)^2$$

or, multiplying by 10 to get the power loss in decibels,

$$\text{Loss in db} = 970 \frac{R_0^{1/2} D}{\lambda^{3/2} V} + 343 \left(\frac{r_0}{\lambda}\right)^2$$

which reduces to

$$\text{Loss in db} \simeq 3.8 \times 10^6 \frac{D}{\lambda^{3/2} V} + 3.4 \times 10^2 \left(\frac{r_0}{\lambda}\right)^2 \quad (8-25)$$

when we select $R_0 = 1.5 \times 10^6$ m as a typical average value for the range.

In calculating the total loss we shall use Eq. (8-18) for r_0 and Eq. (8-21) for D . We shall also take $V = 7 \times 10^4$ m/sec. The contribution from

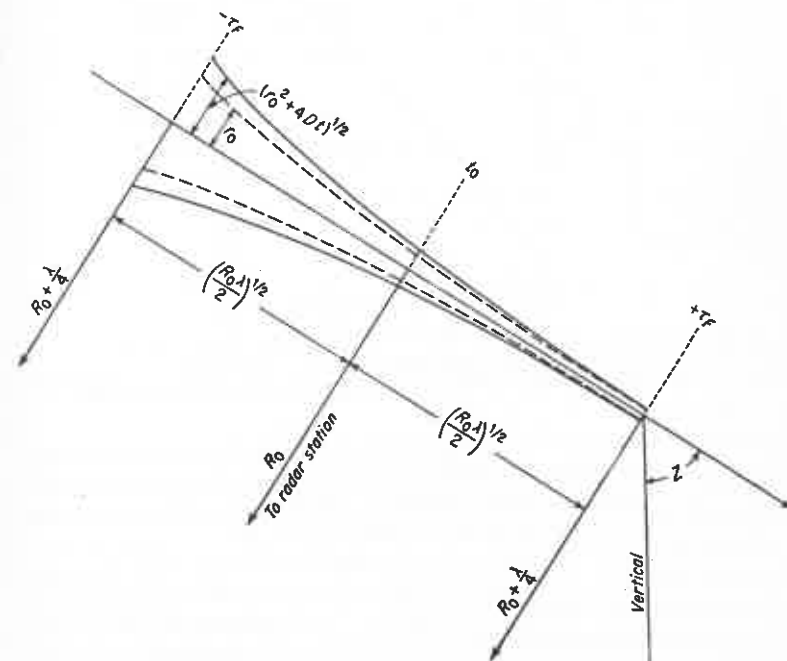


FIG. 8-6. Model of a meteor trail at time τ_F after t_0 , where $\pm\tau_F = (R_0\lambda/2)^{1/2}/V$ defines one-half of the first Fresnel zone of the trail. The initial radius is shown as the dashed line, while the effective radius due to diffusion is indicated by the solid-line envelope.

the D/V term would naturally be greater if we used a smaller value for V , but in evaluating an over-all ceiling, a large V value is more appropriate because high-velocity meteors ionize at greater heights than low-velocity meteors. We shall set an arbitrary level of 40 db for the total power loss as delineating the limit for detection of underdense-type echoes. For this particular value of the total power loss we find that the r_0 term dominates for wavelengths longer than 10 m, while the contribution from the D term becomes more important for shorter wavelengths. A casual glance at Eq. (8-25) might suggest that these statements should be interchanged, since the contribution from the r_0 term varies as λ^{-2} while that from the

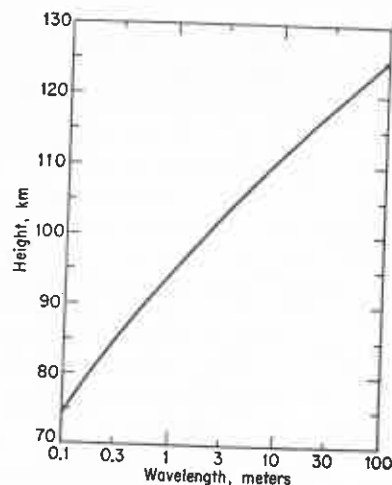


FIG. 8-7. The theoretical underdense echo ceiling, I. The 40-db contour of echo-power attenuation is plotted as a function of height against radio wavelength.

D term varies only as $\lambda^{-3/2}$; this would indeed be right if we were calculating the total loss at a constant height. However, when H is varied in order to maintain a constant 40-db level of the total loss, it will be found that the statements are correct as they stand.

Figure 8-7 shows the variation of ceiling height with wavelength for this arbitrary total power loss of 40 db. One can easily compute curves for other arbitrary values of power loss. It will be found that the cutoff becomes extremely rapid with increasing height: for example, at $\lambda = 10$ m, a height increase of about 2 km results in a total attenuation of 80 db, a further increase of 2 km produces a total loss of 160 db, and so on.

Figure 8-7 should be used with considerable reservation, especially since the absolute heights depend on the adopted model of the atmosphere.

For example, a two-to-one variation in ρ_a or MFP will change the heights by 4 km or so. The Appendix tables show that the present uncertainties in the atmospheric parameters are at least of this order. Furthermore, the ceiling will go up or down with diurnal and annual variations as well as with latitude variations in the parameters. Nevertheless, Fig. 8-7 may be regarded as indicative of the cutoff ceiling to be expected for different wavelengths. The level of 110 km for $\lambda = 9.2$ m is in reasonably good agreement with the observational evidence obtained at Ottawa.

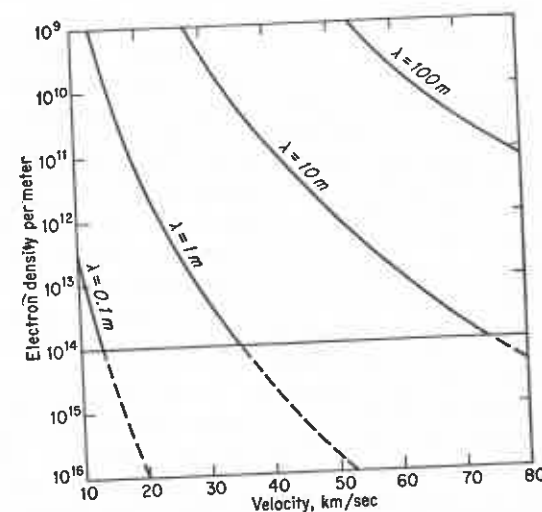


FIG. 8-8. The theoretical underdense echo ceiling, II. The area under a selected wavelength curve shows the range of meteor sizes and velocities for which the attenuation at the point of maximum ionization is equal to or less than 40 db.

It should be emphasized that the ceiling defined here applies to underdense-type echoes. As we shall see in Sec. 8-9, under certain conditions an overdense trail can yield an underdense-type echo, which subjects it to the underdense echo ceiling limitation, but only when those conditions apply.

We may combine the curve of Fig. 8-7 with those in Fig. 7-1 to determine the ceiling conditions for the point of maximum ionization on the trail. In Fig. 8-8, the region below a given wavelength curve shows the combinations of line densities and velocities which may be detected at that wavelength, assuming that a loss of 40 db or less is tolerable at the point of maximum ionization. For example, for $\lambda = 10$ m the graph suggests that a cutoff occurs for $V > 75$ km/sec for underdense trails

with $q = 10^{14}$, the nominal crossover point between underdense and overdense trails. Some meteors in this category will still be detected at points on their trails below the maximum ionization points, but their number will naturally be fewer than the true number. It would appear, therefore, that a strong selection principle is operating to reduce the sensitivity of radio equipment to faint underdense meteors of high velocity. For $V = 40$ km/sec, tenth-magnitude meteors ($q = 10^{12}$) are well under the ceiling, and should be easily detectable with a typical 10-m radar.

One obvious solution to this problem of detecting faint high-velocity meteors is to employ still longer wavelengths. There are practical limitations to the use of wavelengths much longer than about 20 m, say, but the possibilities should certainly be explored. Another solution, not so obvious, is to use very high power and high antenna gain. This will do little toward raising the ceiling for the kind of echoing mechanism that we have been discussing up to now, because of the very great additional attenuation for a small height increment. However, very high power should allow us to ignore the ceiling altogether, because we then should be able to detect the small target presented by the ionization immediately behind the meteor head, before the ion trail expands to such a size that destructive interference attenuates the echo (see Sec. 8-14).

There is evidence that the Ottawa radio equipment was actually not far from realizing the latter objective. An appreciable number of the weaker diffraction echoes showed the typical velocity fluctuations prior to t_0 but no strong enhancement of the echo as it passed t_0 . This behavior is what one would expect from a small target of the moving-ball type. The Millstone Hill radar of the Lincoln Laboratory works on a wavelength of 68 cm and hence should detect practically no underdense trails at all, according to Figs. 7-1 and 8-8. However, through the brute force provided by a peak power of 500 kw and an antenna gain of 37 db, the equipment has detected underdense trail echoes by virtue of this extremely short-trail, or moving-ball, effect. Dr. M. Loewenthal of Lincoln Laboratory, who has carried out some unpublished work on backscatter meteor echoes with this powerful radar, plans to apply the system to the study of forward-scatter meteor echoes. A combination of high power, high antenna gain, and moderately long wavelength should open new vistas to the meteor astronomer.

A third solution to the faint, fast meteor problem is to separate the transmitter and the receiver by distances of the order of 1,000 km. This will be discussed in Chap. 9.

Before we leave this section, let us see what happens as the meteor head in Fig. 8-6 moves on past successive Fresnel zones. We may write $\tau = s/V = x(R_0\lambda)^{1/2}/2V$ in the notation used in Sec. 8-1. Hence the

D term in Eq. (8-25) may be modified to read $3.8 \times 10^4(xD)/(2^{3/2}\lambda^{3/2}V)$. This represents a continuous decrease in echo power as the meteor moves away from t_0 . If D is so small as to be negligible, we obtain the well-known prototype of the diffraction echo (Fig. 8-9, curve A). As the D/V term becomes dominant, the effect is to cause the envelope of the post- t_0 oscillations to approach the zero-power-level axis (see curve B).

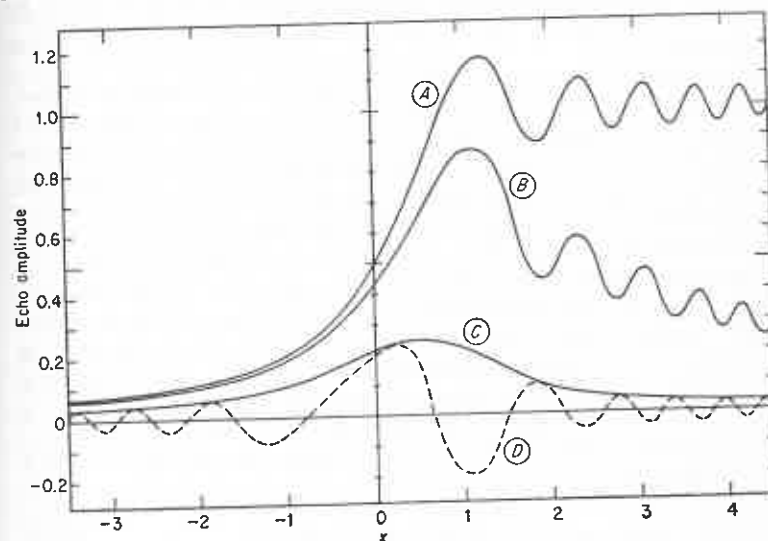


FIG. 8-9. Calculated amplitude-time curves of diffraction echoes from underdense trails. Curve A, no diffusion attenuation, incoherent radar echo; curve B, moderate diffusion attenuation, incoherent radar echo; curve C, severe diffusion attenuation, incoherent radar echo; curve D, same as for curve C, except that coherent detection is used, which brings out the phase variations of the meteor echo.

If the D/V term is large enough, the post- t_0 amplitude oscillations disappear altogether (see curve C), leaving only a smooth rise and fall in signal level at the t_0 point.

The reasons for these effects may be readily understood in qualitative terms from an examination of the Cornu spiral (Fig. 8-3). If there is no attenuation, the base of the amplitude vector remains anchored at the $x_1 = -\infty$ point, in effect, and its tip follows the curve round to $x = +\infty$, producing curve A of Fig. 8-9. With some attenuation at every point, which increases with time, the effective trail length will be finite, and the x_1 point will move in the same direction as the meteor. The echo amplitude will be reduced steadily, but as long as the effective trail length exceeds two or three turns of the upper right-hand spiral the resultant

amplitude will show a few fluctuations (curve *B*), though they will die out as the x_1 point enters the top right-hand spiral. Note particularly that the phase of the Fresnel oscillations is not shifted perceptibly with respect to t_0 (except in the immediate vicinity of t_0); hence no error occurs in the velocity measurements. Once the effective trail length becomes less than one turn of the spiral, the net amplitude will decrease slowly and smoothly without showing any fluctuations, as in curve *C* of Fig. 8-9. The effective length of trail is now so short that we have virtually a moving-ball target.

All of the solid-line curves of Fig. 8-9 are drawn on the assumption that an ordinary noncoherent radar system is used. If we inject a local reference phase, by means of a cw set or a coherent-pulse radar, we can detect the phase variations both before and after t_0 , even when the trail segment has become so short that no amplitude fluctuations at all show on the ordinary radar record. In Fig. 8-9, the same meteor which produced the noncoherent radar record of curve *C* would appear as the dashed-line curve *D* on the record of a cw equipment. (Negative amplitudes are also plotted because the output of a coherent receiver is usually bipolar.) Since the peak echo amplitude of curves *C* and *D* is as much as one-fifth of the unattenuated level shown by curve *A*, there could be a large number of detectable echoes of this kind which will yield velocity measurements with a coherent system but not with a noncoherent system.

8-7. Plasma resonance effects. The dielectric constant of an ionized gas is defined by

$$\kappa = 1 - \frac{N\lambda^2}{\pi} r_e \simeq 1 - 81 \frac{N}{f^2} \quad (8-26)$$

where N is the number of electrons per cubic meter, and the classical electron radius is $r_e = (\sigma_e/4\pi)^{1/2} = 2.8 \times 10^{-15}$ m. λ is the wavelength in meters of the incident radiation, and f is the corresponding frequency in cycles per second.

When an electrically neutral volume of ionized gas (a plasma) is subject to certain boundary conditions, the possibility arises that a resonant oscillation can be induced in the plasma under the influence of the incident radiation, owing to separation of the electrons and ions. In the case of the underdense meteor trail no resonance effects are to be expected if the incident electric vector is parallel to the trail axis, because the ionization is assumed to be both uniform and virtually infinite in length, and hence there are no suitable boundaries. When the electric vector is transverse to the trail, though, it acts across a density gradient. The ions and electrons are separated slightly and the resultant restoring force will induce resonant oscillations. The axial dielectric constant may be highly negative initially. It will increase toward unity as the trail

expands. Kaiser and Closs (1952) have shown that when κ is of the order of -1 , the underdense trail will resonate under transversely polarized excitation. The theoretical maximum enhancement of the transverse echo amplitude relative to the longitudinal amplitude is approximately a factor of two, if a Gaussian radial distribution of ionization is assumed. Collision dampening keeps the resonance from being more pronounced. The transverse and longitudinal reflection coefficients will quickly approach unity again as the trail expands to the order of $r = \lambda/2\pi$, owing to radiation dampening.

This resonance phenomenon was originally predicted by Herlofson (1948b), though his preliminary estimate of the magnitude of the effect was considerably larger than subsequent work proved correct. An early experimental investigation (McKinley, 1951c) showed that if the effect were present the ratio of transverse to longitudinal scattering amplitudes was unlikely to be greater than 4. The reality of the effect and the observational confirmation of the revised theoretical ratio of 2 were demonstrated at Jodrell Bank in a series of careful and ingenious experiments (Closs, Clegg, and Kaiser, 1953; Billam and Browne, 1956). Stanford has also investigated the phenomenon (van Valkenburg, 1954). The phase of the transverse echo vector undergoes a theoretical reversal of 180° as the trail passes through resonance, and this aspect was examined by Greenhow and Neufeld (1956b). The observed phase changes averaged about 40° , with few cases being as large as 180° . The resonance effect usually disappeared about 25 msec after the echo reached its peak amplitude.

As a source of error in velocity measurements, resonance can rarely be considered significant. Observations of decay times can be affected to some extent, if the polarization is not known, unless care is taken to measure the decay-time constant well down on the decay curve, after the resonance effect has disappeared.

8-8. Recombination and attachment. The electrons will eventually recombine with the positive ions to form neutral molecules or atoms. Some electrons may attach themselves to neutral molecules to create negative ions which, though charged, are like the positive ions in being too heavy to oscillate appreciably in the incident field, and hence their scattering coefficient is negligibly small. Both effects can be expected to contribute to the eventual dissipation of the meteor ionization, but the rates at which they operate should be examined to see if they are significant in comparison with diffusion and turbulence.

The differential diffusion equation may be modified to include the recombination effect by subtracting the term $\alpha_e N_e^2$ from the right-hand side of Eq. (8-19). Here, α_e is the electronic recombination coefficient, and N_e is used throughout in the equation to emphasize that the electron

volume density is meant. We shall not go into the analysis here (Kaiser and Greenhow, 1953), since the observations of long-enduring (overdense) echoes have demonstrated that recombination can hardly be a significant factor. The recombination coefficient for meteoric ionization must be of the order of 10^{-16} m³/sec, or even less, in contrast to the generally accepted value of 10^{-14} m³/sec for ionization created at meteoric heights in the atmosphere by solar radiation. Recombination may be neglected in comparison with ambipolar diffusion effects for either underdense or overdense trails.

In considering attachment effects, we subtract a term $\beta_e N_e N_m$ from the right-hand side of Eq. (8-19), where N_m is the density of neutral molecules capable of forming negative ions. In addition, terms should be added for the eventual detachment of the electrons. In the interest of simplicity these terms may be omitted to obtain an upper limit to the effect, if not an exact solution. If we make the plausible assumption that $q = q_0 \exp(-\beta_e N_m t)$ the revised volume density becomes

$$N_e = \frac{q_0}{\pi(4Dt + r_0^2)} e^{-[r^2/(4Dt + r_0^2)] + \beta_e N_m t} \quad (8-27)$$

Molecular oxygen is probably the main constituent of the atmosphere involved in the creation of negative ions. The coefficient β_e is not well determined in this case, but, if $\beta_e N_m t < 1$, Kaiser has shown that the effects of attachment are negligible in the case of the underdense trail. They may also be ignored during the early history of the overdense trail, but as we shall see later, they probably do come into play to limit the very-long-duration echoes.

8-9. The transition from underdense to overdense trails. If the volume density of the electrons is large enough, secondary scattering from electron to electron becomes important. The electrons are no longer independent scatterers, the incident wave does not penetrate the column freely, and the dielectric constant κ is negative throughout an appreciable volume of the gas. These are the conditions for the overdense trail. The transition from underdense to overdense is not sharply delineated. The axial dielectric constant can be highly negative in the underdense trail, but this does not mean that total reflection necessarily occurs. The wave can still penetrate the narrow underdense column, despite the negative κ , though with some loss of strength. Even in metals of high conductivity there is always a certain "skin depth" of penetration of the incident wave, defined as the depth at which the amplitude of the electric vector has fallen to $1/\epsilon$ of the surface amplitude. The important difference between metals and low-density ionized gases is that the conductivity (as expressed in the wave equation) is real for metals and imaginary for ionized gases. In the latter case a cumulative

phase angle is built up by the ray as it passes through the ionized region, since the phase velocity here is greater than the group velocity of light. The greater the density, the greater the total phase angle, relative to free-space propagation. The amount of phase angle which can be tolerated before we define the trail as overdense seems to be pretty much a matter of opinion.

By definition, the critical density N_c is that for which $\kappa = 0$ in Eq. (8-26). As the column expands there will be a critical radius r_c within which $\kappa \leq 0$. For some value of q which marks the transition between underdense and overdense trails this radius will bound a volume both large enough and dense enough to attenuate the wave by $1/\epsilon$. If we put $\kappa = 0$ and use Eq. (8-20) we have

$$N_c = \frac{\pi}{\lambda^2 r_c} = \frac{q}{\pi(4Dt + r_0^2)} e^{-[r_c^2/(4Dt + r_0^2)]} \quad (8-28)$$

Our interest in the problem begins at the moment the trail is created with an initial radius r_0 ; that is, $t = 0$ when $r = r_0$. The transitional value of q is found by setting $r_c^2 = 4Dt + r_0^2 = \lambda^2/4\pi^2$, whence

$$q_{tr} = \frac{\epsilon}{4r_c} \simeq 2.4 \times 10^{14} \text{ electrons/m} \quad (8-29)$$

This is in agreement with the expression derived by Kaiser and Closs (1952). The Stanford group (Manning and Eshleman, 1959), with slightly different but equally acceptable reasoning, define the transition point by

$$q_{tr} = \frac{1}{\pi r_c} \simeq 1.1 \times 10^{14} \text{ electrons/m} \quad (8-30)$$

Note that q_{tr} is a line density. A trail with a line density $q > q_{tr}$ is defined as an overdense trail, and it may normally be expected to yield the overdense-type echo described by Eqs. (8-34) and (8-35) of Sec. 8-10. However, if the wavelength is short or the height is great, the echo is received from an overdense trail may be of the underdense type. This is because we are considering that the echo observations begin only after the trail has reached its initial radius, and under either of the above conditions the volume density on the trail axis, at the instant that r_0 is attained, may be below the requirements for an overdense-type echo. ★★

The condition for an overdense-type echo to be obtained from an overdense trail is that the overdense echo duration T_{oe} [see Eq. (8-34) of Sec. 8-10] shall be greater than the time which would be taken for the trail to reach r_0 , were ordinary diffusion the only factor involved (Manning, 1958). From Eqs. (8-34) and (8-30) we have

$$T_{oe} = \frac{q\lambda^2 r_0}{4\pi^2 D} = \frac{\lambda^2 q}{4\pi^3 D q_{tr}} \quad (8-31)$$

This is to be equal to or greater than the time $r_0^2/4D$ [cf. Eq. (8-28)]. Hence

$$q \geq \frac{\pi^2}{r_0} \left(\frac{r_0}{\lambda} \right)^2 = \pi^2 \left(\frac{r_0}{\lambda} \right)^2 q_{tr} \quad (8-32)$$

If the line density q is below the limiting value given by Eq. (8-32) only an underdense-type echo will be received, even though $q > q_{tr}$. The r_0 ceiling effect of Sec. 8-6 now comes into play, and this underdense-type echo will be attenuated accordingly. Figure 8-10 is a graphical representation of Eq. (8-32) where Eq. (8-18) has been used to establish the connection between H and r_0 .

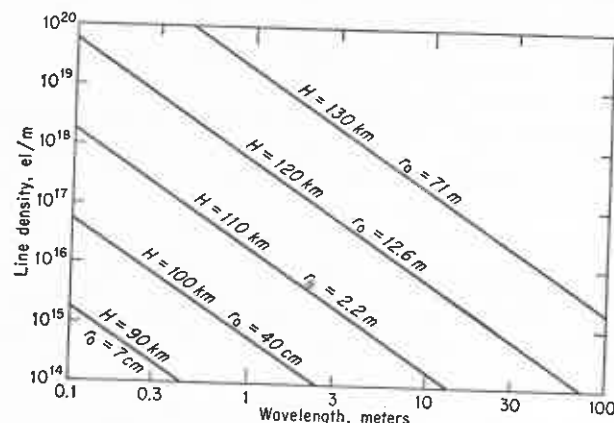


FIG. 8-10. The minimum line density required to produce an overdense-type echo, plotted against wavelength for selected values of H or r_0 .

Various refinements to the approximate theory of this section have been proposed from time to time (for example, Brysk, 1959). The mathematical subtleties of some of these refinements put them beyond the scope of this book—furthermore, not all applied mathematicians in the field are agreed on their validity or usefulness.

8-10. The overdense trail. The basic conditions for the overdense trail and the threshold value of the line density have been discussed in the previous section. As the trail expands we define the effective trail radius as the boundary of the cylinder within which $\kappa < 0$. This radius follows immediately from Eq. (8-28).

$$\begin{aligned} r_c^2 &= (4Dt + r_0^2) \log_e \frac{q}{\pi N_c (4Dt + r_0^2)} \\ &= (4Dt + r_0^2) \log_e \frac{q \lambda^2 r_c}{\pi^2 (4Dt + r_0^2)} \end{aligned} \quad (8-33)$$

The radius grows to a maximum value $r_{c(\max)}$, which is found by differentiating Eq. (8-33) with respect to time and equating to zero. The square of the maximum radius is

$$r_{c(\max)}^2 = \frac{q}{\pi \epsilon N_c} = \frac{q \lambda^2 r_c}{\pi^2 \epsilon}$$

which is independent of r_0 . (However, if $r_{c(\max)}$ should turn out to be less than r_0 , an underdense-type echo only will be obtained after r_0 is reached.) This maximum occurs at time

$$t_{(\max)} = \frac{1}{4D\epsilon} \left(\frac{q}{\pi N_c} - r_0^2 \right)$$

The curve of r_c versus time is fairly flat at the maximum, but eventually the contraction accelerates rapidly. The trail becomes subcritical everywhere at a time T_{ov} (for overdense) which is determined by putting $r_c = 0$ in Eq. (8-28) (ignoring the obvious and trivial alternate solution $t = 0$). This gives us the *overdense echo duration*

$$T_{ov} = \frac{q}{4\pi N_c D} - \frac{r_0^2}{4D} \approx \frac{q}{4\pi N_c D} = \frac{q \lambda^2 r_c}{4\pi^2 D} \quad (8-34)$$

or

$$T_{ov} \approx 7 \times 10^{-17} \frac{\lambda^2 q}{D} \text{ sec}$$

for D in square meters per second, λ in meters, and q in electrons per meter. Except for trails at great heights, or trails for which the density just exceeds the transitional value, we have $q/\pi N_c \gg r_0^2$; hence the r_0 term is customarily omitted when discussing the typical overdense trail. The *echo duration* of the overdense trail, T_{ov} , should be clearly distinguished from T_{un} , the *decay time* of the underdense trail [Eq. (8-24)].

The equivalent echoing area of a long metallic column of radius $r \gg \lambda$ is $\sigma_{cylinder} = \pi R_0 r$ [Eq. (2-17)] for spherical waves incident upon it from a source at a perpendicular distance R_0 . We may regard r_c as the equivalent reflecting boundary of the ionized column, though this is not strictly accurate because the wave will penetrate to some extent inside r_c before total refraction takes place. Furthermore, some refractive effects will be felt before the wave reaches r_c , which will be looked into later. For the moment let us consider that the maximum target area occurs when $r_c = r_{c(\max)}$. Thus $\pi R_0 r_{c(\max)} = R_0 \lambda (q r_c / \epsilon)^{1/2}$. The power flux incident upon the target is $P_T G / 4\pi R_0^2$. The echo-power flux will be attenuated by the same distance factor, $1/4\pi R_0^2$, when the wave returns to the station. The collecting area of the receiving antenna is $G \lambda^2 / 4\pi$. The echo power delivered to the receiver from the overdense trail is obtained

by multiplying these four quantities:

$$P_R = \frac{P_T G^2}{64\pi^3 c^{3/2}} \left(\frac{\lambda}{R_0}\right)^3 r_e^{1/2} q^{1/2}$$

$$= 1.6 \times 10^{-11} P_T G^2 \left(\frac{\lambda}{R_0}\right)^3 q^{1/2} \quad \text{watts} \quad (8-35)$$

Here, $P_R \sim q^{1/2}$, in contrast to the variation of P_R as q^2 in the underdense case [Eq. (8-7)]. Because the received power is relatively insensitive to line density in Eq. (8-35), it is often better to use Eq. (8-34) to determine q for the overdense trail, provided that D is known. This has the further advantage that all of the equipment parameters except λ may be neglected to a first approximation. In passing, it may be noted that the echo power from the underdense trail equals that from the overdense trail when $1/q = r_e^{1/2} (2\pi)^{3/2}$ or $q = 0.75 \times 10^{14}$ electrons/m, which could be said to define yet another nominal value for q_{tr} (Hines and Forsyth, 1957).

In the above discussion the reader will have noticed that we began with a fully fledged overdense trail and skipped the formative stages. If we were to go back to the beginning of the trail history, we ought to arrive at expressions for the diffraction echo which are similar to those deduced in Secs. 8-2 and 8-3—for velocity-measuring purposes they should be effectively the same, in order to conform with observation. An analysis of the formative stage of the overdense trail would not really be redundant, since a different scattering mechanism is involved, but it is very difficult to handle and has not yet been done for the general case. Particular solutions for line densities near q_{tr} have been obtained, but only by long and tedious numerical integrations (Kaiser and Closs, 1952; Keitel, 1955).

Let us now examine the time variations of the overdense echo beginning at the moment r_0 is reached [note that Eq. (8-35) gives only the maximum echo level]. The echo power at any moment will be proportional to r_e , and the time variation of r_e may be found from Eq. (8-33). In Fig. 8-11 the solid curves show the overdense P_R versus time for a number of representative values of q . For convenience we have selected the following typical radar parameters: $P_T = 10^5$ watts, $G^2 = 10$, $\lambda = 10$ m, $R = 150$ km, and receiver noise level 10^{-14} watt. Logarithmic scales are used, after the example of Greenhow (1952a), to keep the diagram within reasonable bounds. All the curves are computed for $H = 93$ km for which $r_{0(\text{ion})} = 12$ cm [Eq. (8-18)] and $D = 4$ m²/sec [Eq. (8-21)]. Theoretically, the initial echo powers from the majority of the overdense trails will lie within a very narrow range—between 35 and 40 db above noise in this case. Few of them should increase more than 10 db during their lifetimes. The logarithmic time scale exaggerates the sharp decrease in echo strength as the echo terminates, but even on a linear scale it would be evident that the echo rapidly drops below noise. When the axial

volume density falls below the critical value, the power level decreases according to the underdense decay curve (see dashed curve for $q = 2 \times 10^{14}$). For long durations this correction to the echo duration is entirely negligible.

In practice, the very-long-duration echoes do not appear either to last as long as the solid-line curves of Fig. 8-11 would suggest, nor do

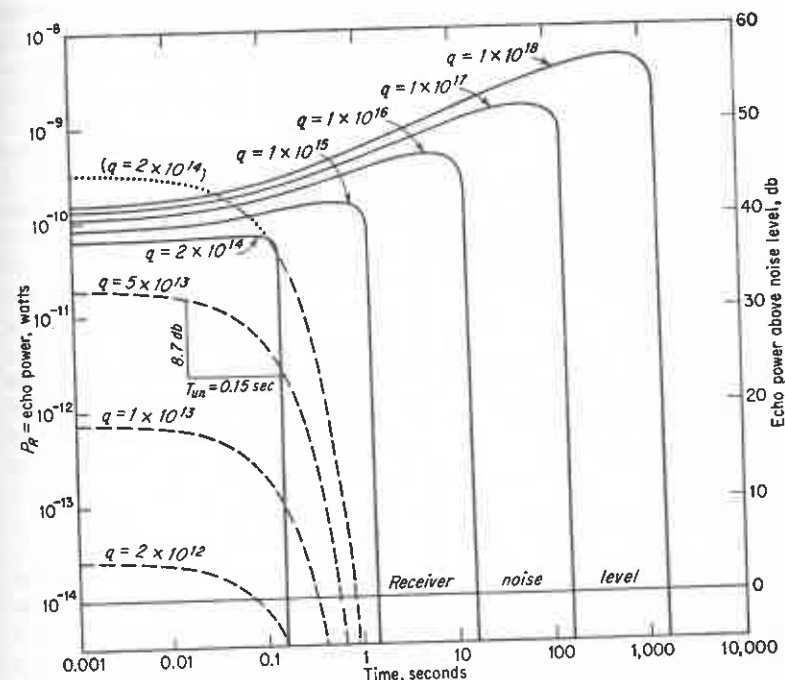


FIG. 8-11. Theoretical variation of echo power with time, based on typical radar parameters as outlined in the text. A diffusion coefficient of 4 m²/sec is assumed; for other values of D multiply the time scale by $4/D$. Solid curves, overdense trails; dashed curves, underdense trails; dotted curve, hypothetical underdense curve for $q = 2 \times 10^{14}$.

they drop off so rapidly. Suppose we were to look at the same meteor with two radar systems which differ in sensitivity by x db, where $x = 10 \log_{10} P_R(2) - 10 \log_{10} P_R(1)$. Here, $P_R(1)$ and $P_R(2)$ are the respective echo powers given by Eq. (8-35). The observed echo durations $T(1)$ and $T(2)$ can be connected by an empirical relation, deduced from simultaneous observations made with a 200-kw radar and a 100-watt radar (McKinley, 1953a). This is shown graphically in Fig. 8-12, where

the ordinate represents the correction to be added to $\log_{10} T(1)$ to yield $\log_{10} T(2)$. The graph is at best an empirical approximation—the implications that a 940-sec-duration echo will be the same duration on any wavelength and that no longer durations ever occur are not really correct—the lines probably should approach the axis asymptotically.

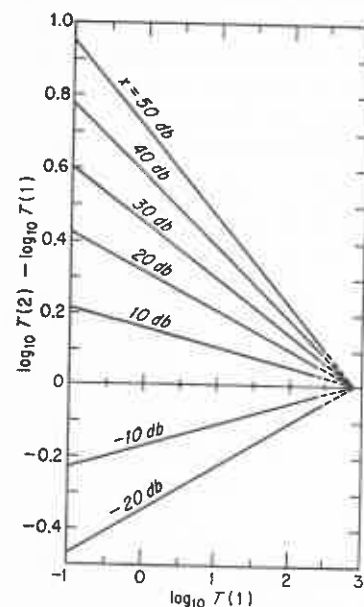


FIG. 8-12. Empirical curves showing the correction to be added to $\log_{10} T(1)$ to yield $\log_{10} T(2)$, when $T(1)$ and $T(2)$ are the observed echo durations on two radar systems differing in sensitivity by x db.

by an arbitrary selection of a tolerable phase angle—we have to accept the total phase angle as it is actually computed at a given instant.

This calculation is not easy, and Manning has carried it out by a form of numerical integration. To summarize his results we may say that the maximum echo power received from the overdense trail will be 70 per cent of the power given by the overdense formula, Eq. (8-35). This quantitative correction to the echo power may not seem of great practical importance in view of the many ill-determined variables which can affect the observed power level much more seriously. From the philosophical viewpoint, though, it makes the story more complete. Furthermore,

Independent confirmation of this general behavior of duration with wavelength was provided by an extensive series of forward-scatter experiments (Forsyth and Vogan, 1956).

Two reasons may be advanced for the difference between the observed and predicted echo-duration curves. First, atmospheric wind motions and attachment of electrons to neutral particles are both effective in shortening the durations of very long echoes—this will be discussed in Sec. 8-12. Second, we have so far neglected to consider the refractive effects of the underdense ionization outside the radius r_c . These can be of appreciable significance, as both Manning (1953) and Kaiser and Closs (1952) have pointed out. To calculate this effect we take the line integral of the refractive index to arrive at a modified formula for the transitional density. The boundary conditions are now established by the time of variation of r_c and the density gradient outside r_c and not

Manning showed that the maximum radius is reached more quickly when refraction is considered, and dies away more slowly, though vanishing at the same time as before [Eq. (8-34)]. This means that the observed durations will depend more on the system sensitivity. Refraction outside the r_c radius can be rather more important in the case of oblique incidence (forward-scatter), when both direction and amplitude are affected.

Let us return for a moment to Fig. 8-11 to discuss the dashed curves. These are plotted for selected q 's, using the underdense trail formulas, Eqs. (8-7) and (8-23). The radar parameters are the same as the ones adopted above for the overdense echoes. In Eq. (8-23) the initial attenuation due to the term $\exp(-8\pi^2 r_0^2/\lambda^2)$ is negligible here if we assume $r_0 = 12$ cm. The decay time, computed from Eq. (8-24) to be $T_{un} = 0.15$ sec, may also be found from any of the dashed curves by marking off an ordinate interval of 8.7 db anywhere on the curve and measuring the corresponding time interval. The minimum detectable signal, with our particular radar, should be obtained from a meteor with q slightly less than 2×10^{12} electrons/m, which corresponds roughly to a visual magnitude of +10.

As we remarked in Sec. 8-9, under certain conditions involving great echo heights or short radio wavelengths only an underdense-type echo can be obtained from an overdense trail (if we assume that our observation begins at the moment r_0 is reached). In Fig. 8-11 the dotted part of the hypothetical underdense echo curve for $q = 2 \times 10^{14}$ electrons/m is not observed. If, from our observation of the decay-type echo represented by the dashed part of this curve, we were to assume that the trail was underdense, we would underestimate the true value of q unless we were able to extrapolate the curve backward along the dotted line. The effect is to lop the top off the assumed echo curve (Forsyth, 1958). If we recognize the echo as due to an overdense trail, the correct value of q can be obtained from the overdense-echo formula. In practice, wind motions and other effects can greatly confuse the issue—in the broad transitional region many echoes of one type can successfully masquerade as members of the other group.

8-11. Enduring echo phenomena. The fading effects observed in long-enduring echoes from overdense trails are explicable in terms of interference between the waves scattered from two or more major echoing points on the trail. Two different mechanisms have been proposed to account for the creation of these discrete scattering centers, either of which can be fitted equally well to the majority of the observations. In certain selected examples, though, it is clear that the smooth-trail "glint" theory (Ellyett, 1950; Greenhow, 1952a; Manning, 1959c) is more appropriate. In other cases the glint theory will not explain observations which the rough-trail "blob" theory (McKinley and Millman, 1949a;

McNamara and McKinley, 1959) can account for quite readily. We shall now examine these two basic hypotheses in qualitative terms.

The glint theory assumes that the ionization is produced continuously and smoothly along the trail. The linear density of electrons rises to a maximum and then decreases, according to the theoretical ionization curve (Sec. 7-6). Wind shears in the upper atmosphere distort the trail more and more as time goes on (see Fig. 8-13). Whenever the tangent to a curved section of the trail becomes normal to the line of sight, an increase in echo power will be obtained from the immediate neighborhood of that point. In effect, a secondary R_0 point has been created on the trail, on either side of which a new first Fresnel zone is formed. The curvature of the trail shortens the zone lengths. Manning (1959c) has

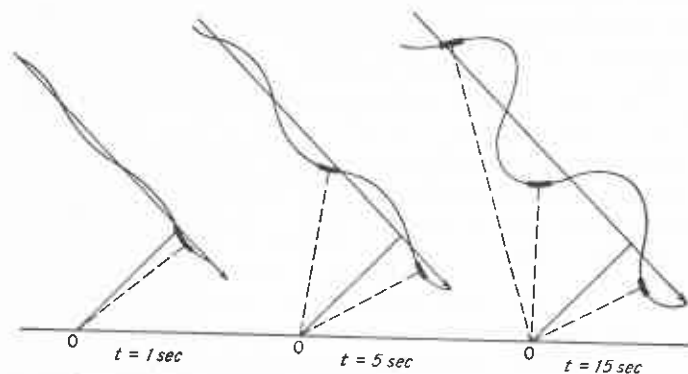


FIG. 8-13. The glint theory, showing how additional reflecting points appear on the trail as distortion progresses.

computed the theoretical delay in the times of appearance of echoes from various parts of the trail. Figure 8-14, *solid curve*, shows this theoretical delay in terms of α , the elongation from the radiant ($\alpha = 0$ is the radiant point; $\alpha = 90^\circ$ is the t_0 point). This delay is roughly proportional to $|\cos \alpha|$, though statistical allowance should be made for the observed fact that wind shears are greater at higher heights. Superposed on Fig. 8-14 are the mean observed delays obtained from visual-radar correlations done many years ago at Ottawa (McKinley and Millman, 1949a). The agreement with Manning's theoretical curve is very good, in view of the wide spread in the original experimental data. For example, in the interval $0 < \alpha < 40^\circ$, the observed delays ranged from zero to over 30 sec—the value of 10 sec shown in Fig. 8-14 is a purely nominal mean.

The blob theory has as its fundamental postulate the requirement that the ionization is not created smoothly and continuously along the trail.

The trail is rough in the sense that the line density of ionization varies significantly over relatively short distances along the path (see Fig. 8-15). As time goes on, the less dense portions become underdense and little or no echo power is received from them. The denser portions grow in size

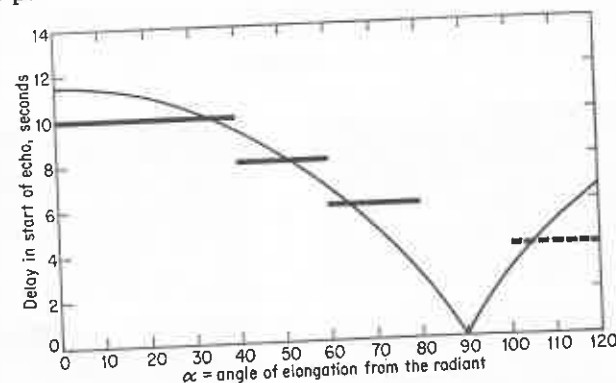


FIG. 8-14. Theoretical and observed delays in the appearance of the echo in terms of α , the elongation of the given point on the trail with respect to the meteor radiant. *Solid curve*, the glint theory (after Manning, 1959c); *horizontal lines*, observed delays for selected visual-radar correlations.

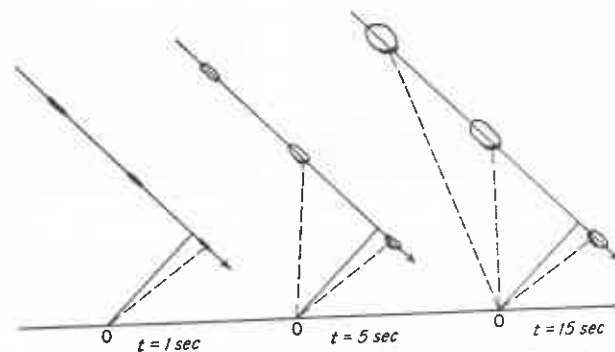


FIG. 8-15. The blob theory, showing how additional targets appear on the trail as the more densely ionized parts grow in volume due to diffusion.

as the electrons diffuse. The ratio of the radius to the length of a given segment increases until ultimately the blob may approach roughly spherical proportions, while remaining overdense. The effective echoing area of a blob at any time will be proportional to the product of the principal radii of curvature at the point where the line of sight intersects the critical density surface. For a given blob at any instant this product will be

greatest if $\alpha = 90^\circ$ and least if the blob is viewed end-on at $\alpha = 0^\circ$. On this theory the delay in the appearance of the echo is also roughly proportional to $|\cos \alpha|$.

A further factor enters, though, because the rate of growth of the blob target increases with D , and D always decreases as α increases. One can select a typical set of conditions under which the calculated curve will display an excellent fit with the selected observational data given in

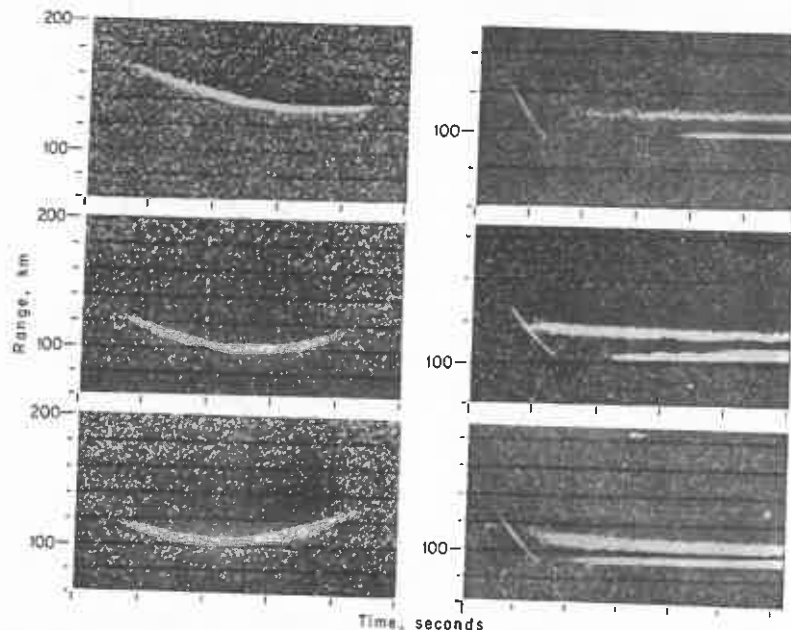


Fig. 8-16. *Left*, head echoes from three meteors which left no enduring ionization; *right*, head echoes from three meteors—enduring ionization has appeared in each case, at two or more quasi-stationary ranges, and with delays up to two or three seconds. (Note that these are single-station records of six *different* meteors, not three-station records of two meteors, despite the seeming similarity to Fig. 4-8.)

Fig. 8-14. One may also choose another set of conditions which will actually show a marked increase in delay as α approaches 90° . To take an extreme case consider a meteor coming nearly head-on to the station from a radiant high in the sky. The diffusion coefficient will decrease by one or two orders of magnitude as the meteor falls, but α will not increase very much, and the diffusion factor will overrule the $\cos \alpha$ factor. This prediction is in agreement with many of the observational records (see Fig. 8-16, *right*). These illustrations show that meteors in the visual

range—not necessarily the brighter ones only—can produce enduring echoes at points on the trail far removed from the t_0 point. The delays are often far too short to be explained by the glint theory. In many cases there may be no measurable delay at all relative to the head echo (see Fig. 8-17). The discrete echoing areas have been formed nearly instantaneously. The discreteness of the echoing sources follows from the assumption of an initially rough trail.

In the above discussion we have tacitly assumed, first, that the glint theory was based on a smooth trail and relied on wind shears alone to produce the glints, and second, that the blob theory demanded a rough trail but needed no wind shears. However, in both cases the restrictions are unnecessary. The smooth trail is not really essential to the glint theory, and the inclusion of wind shears will improve rather than spoil the blob theory. The fragmentation hypothesis, which states that meteoroids break up into many fine particles as they encounter the air resistance, is additional evidence for suggesting that the majority of meteor trails are rough, in the visual range of meteor sizes at least. This is probably the most important cause, though other physical processes may also be visualized to account for the initial roughness of meteor trail ionization, such as rotation of a nonspherical meteoroid—this could lead to a regularly repeated variation along the path which is apparent in many of the individual records.

The equivalent echoing areas of the *curved* glint segment and of the blob are both of the same form, namely, $\pi r_1 r_2$ (Sec. 2-10), and are both of the same order of magnitude. In neither case has the echo much depth in range: practically all of the echo is received from a small area of that part of the surface facing the line of sight, no matter what the shape of the rest of the body may be. As to the variation of echo delay time with wavelength, we may note that the critical radius increases with wavelength [Eq. (8-33)]. Either the glint segment or the blob will therefore appear as a larger target on a longer wavelength, and should therefore return a detectable echo sooner—which is in accord with the observations. Occasionally, a radar echo will change fairly rapidly and continuously in range as the glint point follows the convolutions of the trails. However, most of the discrete enduring echoes show no change of range at all, other than a small translational motion which can be attributed to ordinary wind drifts. Furthermore, many discrete echoes show no aspect sensitivity (Fig. 8-17), although they appear immediately after the passage of the meteor. This behavior is somewhat easier to explain if the trail ionization varies markedly along its length.

Analysis of echo fading rates with cw or long-pulse radar equipments shows that on the average a delay of 0.4 sec occurs before fading begins, that is, before a second reflecting center is formed. The spectrum of the

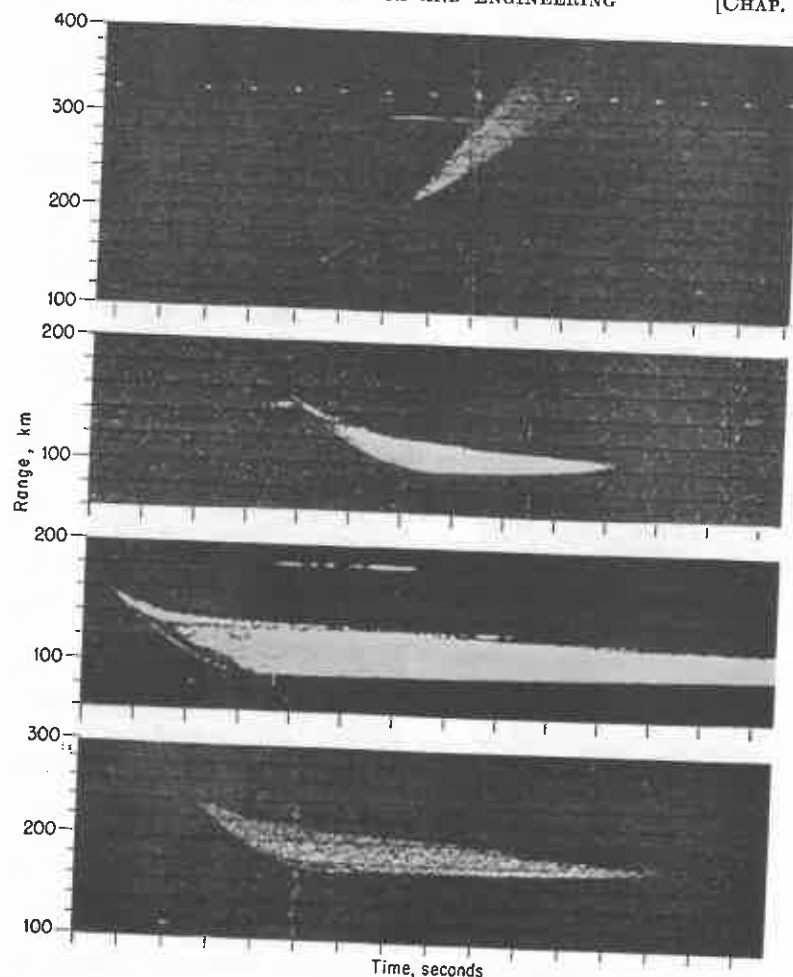


FIG. 8-17. Radar echoes from meteors. In each case the enduring ionization is practically continuous from all parts of the trail. The echo duration at any point of the trail is dependent both on the height and the rate of production of ionization at that point.

fading frequencies becomes more complicated as additional centers come into play. Finally, as the echoes die away from the parts at either end of the trail, the spectrum narrows down again to a single component. On the average, the longest echo endurance for a 40-km/sec meteor will be found at a height of 93 km. Above this height, diffusion quickly thins

the ionization to underdense values while at lower altitudes some other processes, probably turbulence and attachment, act to shorten the duration. One can interpret the fading spectra in terms of turbulent conditions in the upper atmosphere. We shall not pursue this particular line of research further here because the meteor evidence, although important, is only a small part of all the available data in the field of ionospheric motions.

To sum up, we may say that neither theory alone is sufficient to explain all the phenomena adequately. If we combine the wind-shear action with the rough-trail hypothesis, a satisfactory compromise is reached which can cope, in qualitative terms at least, with all the observational evidence advanced so far.

8-12. Echo duration limited by attachment. Because of the initial aspect sensitivity of most meteor trails it is difficult to get a clear conception, from single-station observations at least, of the variation in line density of ionization along the entire trail. Exceptions to this general statement may be found in certain radar-echo records like Figs. 8-17 and 8-18. The implication of these records is strong that we have obtained here a fairly realistic picture of the typical variation of echo duration at every point on the trail, free of any aspect-sensitivity considerations. Figure 8-19 is a height-duration plot of the three-station meteor echo shown in Fig. 4-8. The observed duration at any height level is probably independent of the viewing aspect in this case too.

The problem arises in interpreting the echo duration at any given point in terms of the initial ionization produced there. For example, Fig. 8-18 shows a steadily increasing echo duration as the height along the path decreased. This can quite clearly be attributed both to the increasing rate of production of ions and to the decreasing value of D as the meteor descended. At some height a maximum duration was attained, below which the duration rapidly fell off to zero, but in the example given this cutoff cannot be due to a decrease in the rate of production of ionization. The head echo continued to be very strong, and it is unlikely that the point of maximum ionization was reached much before the meteor arrived at t_0 , which is well below the height where any appreciable duration of the echo can be perceived. Some other physical process must be operating to reduce the echo duration so severely. The same "floor" effect is noticed in the photographic evidence of meteor trains (see Sec. 5-8).

Diffusion at heights below about 90 km is so slow that it cannot be considered as an effective agent. If we relied on diffusion alone to limit the echo duration we would find, for example, that Eq. (8-34) predicts a duration of over 3 hr on a 10-m radar for a -5 magnitude meteor moving at 40 km/sec. This is based on the assumption that the maximum duration would occur at the height of maximum ionization, which in this case

is about 80 km (Fig. 7-1). In practice, though, the average observed duration of the same meteor would be more like 5 min, and the observed height of the maximum echo endurance would be about 93 km.

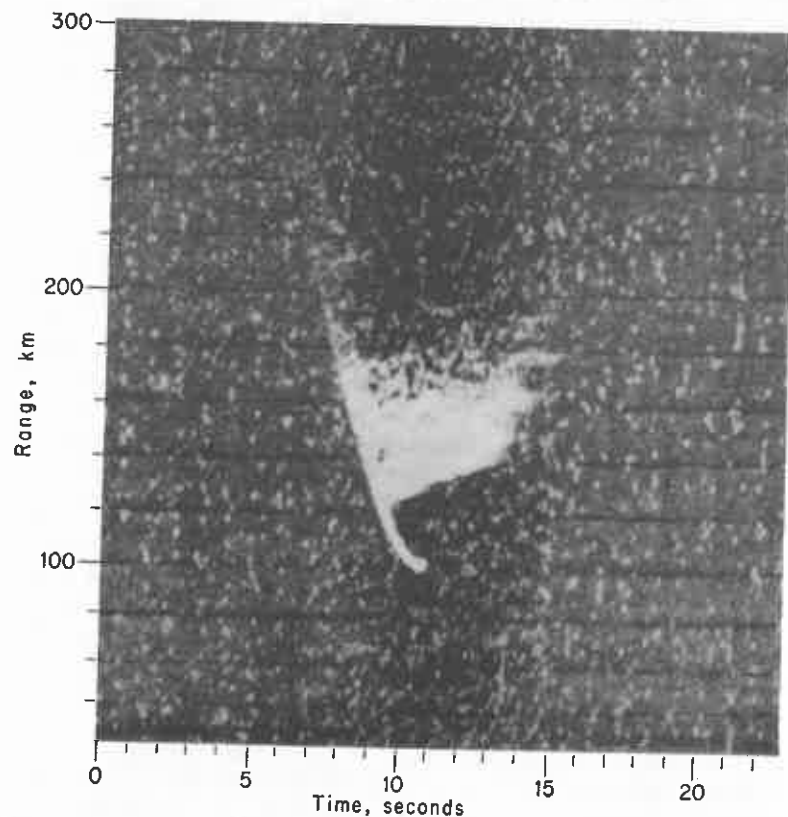


FIG. 8-18. A radar echo from a meteor which has passed right through the meteoric region before burning up. Note also the enhanced noise spots which appear on the record at all ranges while the enduring ionization persists. This effect is discussed in Sec. 10-1.

Recombination of the positive ions and the electrons has received serious consideration but has been rejected as a plausible cause. As Manning and Eshleman (1959) put it, recombination has been demonstrated to have negligible effect during the early history of the trail, and since the time constant of the recombination process increases with the age of the trail, it may therefore be considered entirely negligible during the later history.

Turbulence in the atmosphere will in time stir up the trail and redistribute the ionization. Greenhow and Neufeld (1959a) find that the turbulence is distinctly anisotropic; the vertical scale of the large eddies is roughly 7 km and the horizontal extent is about 150 km. [Large-scale motions of this order of size may be due to internal gravity waves in the upper atmosphere rather than to turbulence, according to Hines (1959).] The size of the smallest eddies is approximately 50 m. Even a smooth-trail model should eventually become rough by this process. The echo durations will be decreased and the dependence of duration on λ^2 will no longer hold. It has been noted from simultaneous observations made on two wavelengths (Greenhow, 1952a; McKinley, 1953b) that $T \sim \lambda^2$ for

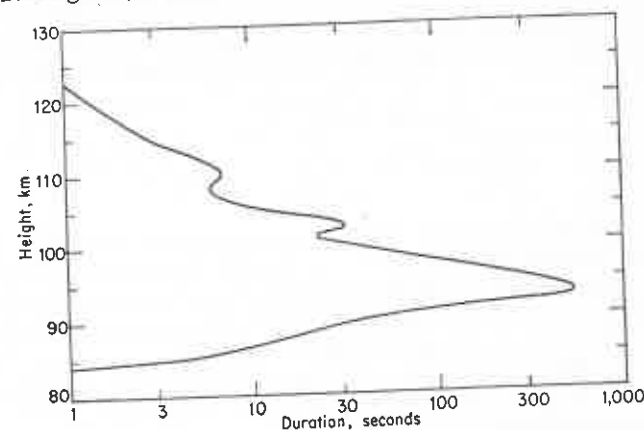


FIG. 8-19. Profile of echo height versus duration, for the three-station meteor echo of Fig. 4-8.

trails with electron densities not greatly in excess of the transitional density, but that the power of λ approaches unity for very-long-duration echoes. The theory of atmospheric turbulence, which is based only in part on the meteor observations, is undergoing continuous revision, and the discussions have occasionally been rather controversial (Booker, 1956, 1958; Booker and Cohen, 1956; Manning and Eshleman, 1957, 1958; Manning, 1959b, 1959c; Greenhow and Neufeld, 1959a, 1959b; Rao, 1959; Rao and Armstrong, 1958; and others). In one series of papers on the mechanics of turbulence and its effect on meteor echoes (included in the list above) both protagonists in the debate cited the same Ottawa observations (McKinley, 1953a, 1954b) in support of their widely differing viewpoints—which left the originator of the data both confused and complimented. Fluid dynamics is too involved a subject to be treated adequately in this book, and we shall not enlarge on its applications here.

The interested reader will find that accepted theories of comparatively recent vintage are being subjected to drastic overhaul in the current literature.

Attachment of the electrons to neutral atoms or molecules has just recently been put forward again as a process which seems quite capable of controlling the durations of very long echoes without greatly affecting the shorter echoes (Davis, Greenhow, and Hall, 1959a, 1959b). We have already obtained the echo duration T_D , based on diffusion alone, by putting $r_c = 0$ in Eq. (8-28). That is, $N_c = q/4\pi DT_D$, since we may neglect the r_0 term for long-duration echoes. Similarly, the duration T_A , which would be observed in the presence of both attachment and diffusion, is found by putting $t = T_A$ and $r = 0$ in Eq. (8-27). Thus, $N_c = (q/4\pi DT_A) \exp(-\beta_e N_m T_A)$. Equating these expressions, since the same limiting value of critical density is involved in both cases, we have

$$T_D = T_A e^{\beta_e N_m T_A}$$

or

$$\beta_e N_m = \frac{2.303}{T_A} (\log_{10} T_D - \log_{10} T_A) \quad (8-36)$$

Next, let us examine the echo-duration correlation with visual magnitude for Perseid meteors observed at Ottawa (Millman and McKinley, 1956). The data show very wide spreads in both duration and magnitude—a given duration class may extend over eight or more magnitudes, or a given magnitude class may include durations which vary as much as 1,000:1. If we plot the mean of the logarithms of the durations for each magnitude class, we obtain the dashed curve of Fig. 8-20. Or, if we plot the mean magnitudes for selected duration classes, we arrive at the dotted curve of Fig. 8-20. Obviously, the true relation lies somewhere between these extremes. Empirical error functions have been evolved, based on the observed distributions. With these error functions, corrections have been computed for each of the points shown on Fig. 8-20. The corrected points (not shown) lie very closely about the solid straight line marked "Diffusion," over the magnitude range -3 to $+5$. For the brighter magnitudes the error functions become less reliable because of the scarcity of data. It is obvious, though, that the true curve must dip downward from the straight line and lie between the dotted and dashed curves as M becomes more negative. Even over the magnitude range -3 to $+5$ there is no assurance that the analytical procedure has not masked some small but real deviations from the calculated straight line.

We have computed values of $\log_{10} T_A$ for selected values of $\beta_e N_m$ in Eq. (8-36), taking the $\log_{10} T_D$ values from the extrapolated straight line. The most reasonable fit appears to result when $\beta_e N_m = 0.006$. This is shown in Fig. 8-20 as the solid curved marked "Attachment." Davis,

Greenhow, and Hall employed the Ottawa data somewhat differently to arrive at a value of $\beta_e N_m$ between 0.015 and 0.025. We are inclined to favor a value not exceeding 0.01. If $\beta_e N_m$ is greater than this, the theoretical maximum duration approaches a limit of 200 or 300 sec. Five-minute echoes are not uncommon at $\lambda = 10$ m. A few have been

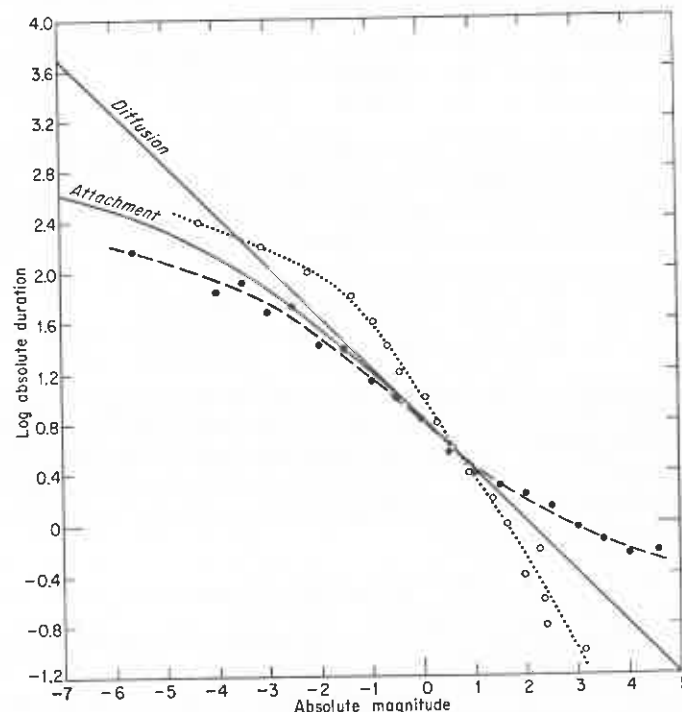


FIG. 8-20. Correlation between echo duration and magnitude of 1,404 Perseid meteors. Solid circles and dashed line, mean of log durations in given magnitude classes; open circles and dotted line, mean of magnitudes in given log duration classes. The straight line marked "Diffusion" is the best fit to the data over $-3 \leq M \leq 5$. The curve marked "Attachment" has been computed as described in the text.

observed with durations of 30 min or longer; they have not occurred often, it is true, though the frequency of their appearance was not at variance with the statistical expectation of extremely bright fireballs (McKinley and Millman, 1953; McKinley, 1954b). During daylight a photo-detachment process may operate to decrease β_e . If so, the enhanced ultraviolet radiation that causes a sudden ionospheric disturbance might further lengthen the durations.

8-13. The radio magnitude relation. In addition to the data on the Perseids ($V = 60$ km/sec) given in the previous section, the Ottawa observations included similar information on 256 Geminids ($V = 36$ km/sec) and on 1,420 nonshower meteors whose average velocity was taken to be about 45 km/sec. The equations below represent the three lines which were fitted to these sets of data.

$$\begin{aligned}\log_{10} T_D &= 0.87 - 0.406M && \text{(Perseids, Fig. 8-20)} \\ \log_{10} T_D &= 1.10 - 0.378M && \text{(nonshower)} \\ \log_{10} T_D &= 1.15 - 0.349M && \text{(Geminids)}\end{aligned}\quad (8-37)$$

In the formula $T_D = 7 \times 10^{-17} \lambda^2 q/D$, we put $\lambda = 9.2$ m; for the Perseids we have $D = 13$ m²/sec at 100 km; for the Geminids, $D = 3.2$ m²/sec at 91 km; and for the nonshower meteors, $D = 5.0$ m²/sec at 94 km [from Fig. 5-10 and Eq. (8-21)]. Eliminating T_D , the above equations convert to

$$\begin{aligned}M_r &= 40.0 - 2.45 \log_{10} q && \text{(Perseids)} \\ M_r &= 42.4 - 2.65 \log_{10} q && \text{(nonshower)} \\ M_r &= 45.5 - 2.87 \log_{10} q && \text{(Geminids)}\end{aligned}\quad (8-38)$$

If the coefficient of $\log_{10} q$ were exactly 2.5, it would imply that a constant ratio exists between I and q for a given meteor velocity, independently of magnitude. On the other hand the relation between ionization and velocity may well differ from that between luminosity and velocity. As a compromise to the three observational equations [Eq. (8-38)] we have evolved the general formula

$$M_r = 36 - 2.5 \log_{10} q + 2.5 \log_{10} V \quad (8-39)$$

which is not a bad fit to all three, if we agree that a common slope of 2.5 is acceptable and that the dependence on velocity can be taken care of by a simple vertical translation of the lines. Hawkins has developed a somewhat similar velocity correction from independent data.

Purely as a pragmatic expedient, and with even less physical justification than we invoked in producing Eq. (8-39), it happens that a very good fit can be obtained for all the lines of Eq. (8-38) by means of the simple expression

$$M_r = \frac{71 - 4.4 \log_{10} q}{\log_{10} V}$$

This we shall not dignify with an equation number.

We note that the three empirical lines in Eq. (8-38) are reasonably well satisfied by $q = 10^{16}$ when $M = 0$. This gives us the link between magnitude and ionization of which we have made frequent use. The simplified form of the radio magnitude equation which we first introduced in

Chap. 2 then follows immediately:

$$M_r = 40 - 2.5 \log_{10} q \quad (2-3)$$

which is fairly satisfactory for $-2 \leq M_r \leq 5$ and average meteor velocities. Equation (8-39) reduces to Eq. (2-3) when $V = 40$ km/sec.

A connection between τ_q and τ_I may be deduced from Eq. (8-39). First, by dividing Eqs. (7-3) and (7-6) we have $I \sim qV\tau_I/\tau_q$. From Eq. (2-2) $M \sim -2.5 \log_{10} I \sim -2.5[\log_{10} q + \log_{10} (V\tau_I/\tau_q)]$. Comparing this with the observational Eq. (8-39), we obtain $\log_{10} V \sim -\log_{10} (V\tau_I/\tau_q)$, or $\tau_q \sim V^2\tau_I$.

Although the differences in the slopes of the observational relations between M and q are small, they are probably quite real, and we have no justification for forcing them into a common mold other than simplicity and convenience. From Fig. 5-11 we have concluded that $H = 93$ km represents the height of maximum echo duration which would be observed if a meteor were to produce ionization at a constant rate all along its path. Above 93 km diffusion is the controlling factor, and below 93 km attachment is possibly the chief agency reducing the durations. The duration of a fifth-magnitude Perseid meteor will therefore be less than that of a fifth-magnitude Geminid meteor, because the maximum ionization of the Perseid occurs at 106 km and that of the Geminid at slightly more than 95 km (Fig. 7-1). On the other hand, a -5th-magnitude Perseid ($q_{\max}$ at 89 km) will have a longer duration than a -5th-magnitude Geminid ($q_{\max}$ at 79 km). Therefore the slope of the Perseid line of Eq. (8-37) should indeed be steeper than that of the Geminid line.

We can easily calculate how far the point of maximum echo duration will lie below the height of maximum ionization, in the region above 93 km where diffusion alone need be considered. From Eq. (8-34), $T/T_{(\max)} = qD_{(\max)}/q_{\max}D = q\rho_a/q_{\max}\rho_{a(\max)}$, if we tentatively assume $D \sim 1/\rho_a$. Thus, from Eq. (7-29) we have

$$\frac{T}{T_{(\max)}} = \frac{9}{4} \left(\frac{\rho_a}{\rho_{a(\max)}} \right)^2 \left(1 - \frac{\rho_a}{3\rho_{a(\max)}} \right)^2 \quad (8-40)$$

If we differentiate T with respect to ρ_a and equate to zero, we obtain immediately $\rho_a = 1.5\rho_{a(\max)}$ for the density at the height of maximum duration T_m . At this point $q = (2^{1/2}/2)q_{\max}$ and $T_m = (8^{1/2}/64)T_{(\max)}$. In terms of height we have $0.075(H_{(\max)} - H) = \log_{10} 1.5$, or $H_{(\max)} - H \simeq 2.3$ km. The height of the longest echo will therefore be 2 or 3 km below the height of maximum ionization (this will vary with the scale height adopted). However, this constant differential cannot hold as $H_{(\max)}$ approaches 93 km. At heights where diffusion alone is operative we should multiply the q 's deduced from the observed T_m 's by the factor $3^{3/2}/7$ to obtain $q_{\max}$ values, for truer correlation with the visual magnitude estimates. This multiplying factor will approach unity at 93 km, and

below this height it will increase again in some manner, though for entirely different reasons. In a qualitative sense this could be interpreted in terms of rotation of the M versus $\log_{10} q$ lines: the amount of rotation would depend on the pivotal point, which in turn depends on the velocity. It would not require much ingenuity to juggle the corrections arising from this *ad hoc* logic to yield an ideal straight line to fit all meteor sizes and velocities. The practical virtue of such a procedure is dubious and the theoretical validity is weak. The true relation between M and $\log_{10} q$ is not likely to be represented by an exact straight line, even over the range $-3 \leq M \leq 5$. Until more information becomes available, though, one may use the simple formula [Eq. (2-3)] as a fair working tool, or the one with the velocity correction [Eq. (8-39)] if more precision is needed. (The reader may use the unnumbered M versus $\log_{10} q$ formula at his own risk!)

In deriving Eq. (8-40) we assumed tentatively that $D \sim 1/\rho_a$, whence $T/T_{(\max)} \approx q\rho_a/q_{\max}\rho_{a(\max)}$. From Eq. (7-24), $\rho_{a(\max)} \sim q_{\max}^{1/2}$; so

$$\frac{T}{T_{(\max)}} = \frac{q^{1/2}}{q_{\max}^{1/2}} \quad (8-41)$$

In practice (see Appendix for an explanation) we have adopted a slightly larger effective scale height for the diffusion equation ($H_D = 0.434/0.067 = 6.5$ km) than for the other atmospheric parameters ($H = 0.434/0.075 = 5.8$ km). Hence, in our empirical relation, the power of q will be slightly less than $1/2$. In fact, combining Eqs. (7-27), (8-21), and (8-34) we have

$$\log_{10} T_{(\max)} = 1.30 \log_{10} q_{\max} - 3.3 \log_{10} V - 2 \log_{10} \lambda - 16.0 \quad (8-42)$$

$\log_{10} T_m$ is obtained by adding $\log_{10} (81/64)$ to Eq. (8-42). Assuming $\log_{10} q \sim \log_{10} I \sim -0.4M$ one might think that the slopes of all the lines of Eq. (8-37) should be $-1.30 \times 0.4 = -0.52$, which they are not. We would therefore be no further ahead if we used Eq. (8-42) as the intermediary between Eqs. (8-37) and (8-38), although the presence of the $\log_{10} V$ term in Eq. (8-42) helps to rationalize the empirical inclusion of $\log_{10} V$ in Eq. (8-39). The basic observational data should be regarded as authoritative, but the necessary processing of the data for comparison with theory is itself an interpretative procedure which involves theoretical considerations and subjective opinions. The problem is intriguing and will bear further examination, both from the experimental and the theoretical viewpoints.

8-14. The head echo. The integral of the field vector given in Eq. (8-3) is quite general and applies all along the trail. In Sec. 8-1 we were interested only in the portion near t_0 , and hence we were able to carry out the integration by making an approximation for R which is valid only near t_0 . The increase in echo strength as the meteor nears t_0 is due to the increasing length of each new Fresnel zone and not to the decrease in R ,

which is negligible. Going to the other extreme, where the meteor is approaching directly toward the station, the converse situation exists: each Fresnel zone is the same length, namely $\lambda/4$, and any over-all increase in echo power will be due chiefly to the decrease in range, if we assume a constant q . In this case we may write R for s in Eq. (8-3). When R is large, the net amplitude due to any two adjoining Fresnel zones is very close to zero. In effect, the last zone to be formed will largely determine the instantaneous magnitude and phase of the total amplitude vector A_R . We therefore need integrate the phase angle $4\pi R/\lambda$ only over the distance $\lambda/4$. Over this small interval the range factor in ΔP_R may be considered constant.

$$A_R(\text{end-on}) = (2\pi \Delta P_R)^{1/2} \cdot q \cdot \frac{1}{2} \int_0^{\lambda/4} \sin \left(2\pi f t - \frac{4\pi R}{\lambda} \right) dR$$

$$\text{whence} \quad P_R(\text{end-on}) = \frac{A_R^2}{2\pi} = \Delta P_R \frac{\lambda^2}{16\pi^2} q^2 \quad (8-43)$$

For any other trail orientation the echo signal will increase smoothly from this limiting asymptotic value to the maximum value obtained when the meteor passes t_0 . The analysis above is equally valid for meteors receding from the station, and if the radar range resolution is adequate, both the approaching and receding echoes will be separable from the strong echo at t_0 . If our end-on trail were rotated 90° so that it crossed the line of sight at range R (with the meteor head well past t_0), the echo power would be $P_R(\text{normal}) = \Delta P_R R \lambda q^2 / 2$, from Eq. (8-6). The ratio of these powers is

$$\frac{P_R(\text{end-on})}{P_R(\text{normal})} = \frac{\lambda}{8\pi^2 R} \quad (8-44)$$

When $\lambda = 10$ m and $R = 150$ km, the power level of the end-on echo will be more than 60 db below $P_R(\text{normal})$. Although weak, these echoes are easily detectable with conventional cw meteor systems. In addition to the stronger echoes which change pitch rapidly near t_0 , many fainter Doppler whistles of nearly constant pitch can be recorded from meteors far from the t_0 point.

A diffraction echo will thus be obtained from every trail as it is being formed, no matter what the trail orientations may be, but the intensity will fall off rapidly on either side of t_0 . Browne and Kaiser (1953) suggested that this simple extension of the diffraction theory to points remote from t_0 would account for the head echoes which are observed on range-time radar records. The major objection to this is that the observed variation in intensity of the head echoes is usually very much less than that predicted by the diffraction theory (McKinley, 1955). The brighter the meteor, the less the variation in strength as the meteor passes t_0 . These echoes probably come from highly overdense trails. It has been sug-

gested (McKinley and Millman, 1949a) that the transient head echo is received from a large cloud of ionization which accompanies the meteor head. The echo from such a moving-ball target will be relatively independent of aspect. For very bright meteors the diffraction echo may be negligible in comparison with the moving-ball echo, even near t_0 . Occasionally, interference effects are observed near t_0 (see Fig. 4-9b) which support the thesis that these two types of echoing mechanism do exist together. Some typical head echoes are shown in Fig. 8-21; others appear in Figs. 8-16 and 8-17.

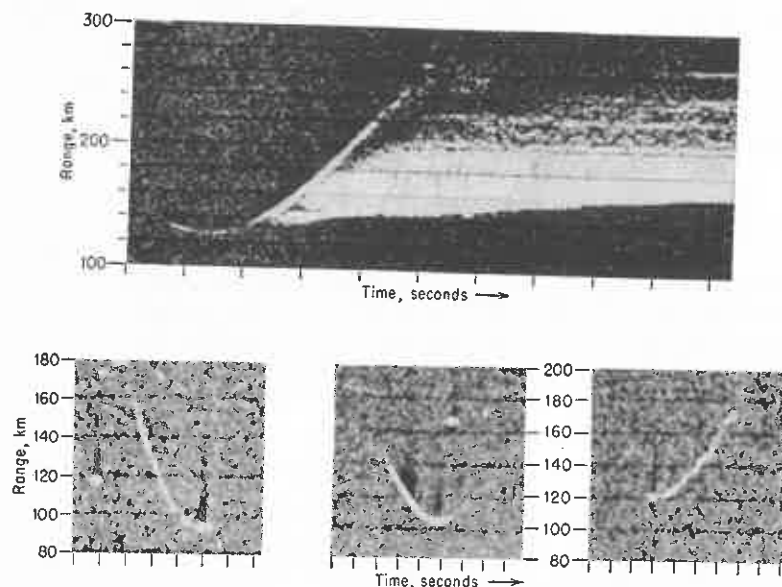


FIG. 8-21. Meteoric head echoes.

It has been difficult to visualize a reasonable physical process which will account not only for the creation of a large ionized cloud around the meteor head but also for its necessarily rapid dissipation, since another striking feature of the head echo is that, as a type, it shows little or no measurable duration at any point of the path. Shock waves have been proposed as a possible ionizing agent, though the theory encounters some problems where the air density is very low—head echoes are observed as high as 130 km. The complete lack of aspect sensitivity of the true head echo (it seems not to matter whether the meteor is approaching or receding from the observer) might be hard to explain in view of the probable conical outline of the ionization created by the shock-wave front.

Another suggestion was offered some time ago (McKinley and Millman, 1949a) that ultraviolet light from the meteor head could ionize the air particles at a considerable distance from the meteor. This was criticized partly on the grounds that the amount of available uv energy seemed inadequate, and partly because the estimated values of the recombination or attachment coefficients of any of the known atmospheric particles were too small to allow for the rapid disappearance of the echo. Cook and Hawkins (1960) have recently concluded that neither objection may be insuperable. According to them, the ionizing radiation from a meteor of magnitude -5 is of the order of 10^7 watts, and the spherical volume thus ionized can have a critical density radius of several hundred meters. Bialecke and Dougal (1958) have shown that molecular nitrogen ions have a recombination coefficient of 9×10^{-13} m³/sec at meteoric heights. Although this value for N_2^+ is considerably larger than earlier work had indicated to be possible, it is nevertheless too small for the present purpose and therefore seems to rule out N_2^+ as the major constituent of the ionized ball. A recombination coefficient of the order of 2×10^{-11} m³/sec is needed to account for the rapid disappearance of the echo, and Cook and Hawkins are of the opinion that the dissociative recombination coefficient of photo-ionized molecular oxygen may meet the requirements adequately. Other questions remain, though, such as a predicted decrease in diameter of the ionized ball as the meteor falls. Owing to the increasing air density the absorption of the uv light should occur within a smaller volume as the height decreases. It may be that this effect is counterbalanced by the increasing light output prior to the point of maximum light. Much more work must be done before a final conclusion can be reached about the validity of the uv light hypothesis.

The term "head echo," as used in this book, is intended to apply to the bright-meteor phenomenon we have just been discussing, where the head-echo amplitude does not vary much along the path. In the broadest sense, of course, every meteor is capable of producing a diffraction-type echo continuously as it moves along its path, though at points remote from t_0 this echo is usually below the level of detection. When D and r_0 are both very large, as is the case for very faint, fast meteors, the first few meters of trail behind the meteor head may be the only part of the trail from which we can hope to get an echo at all. These echoes are also of the moving-ball type, similar to the bright-meteor head echoes, but the mechanism of their production is clearly different. At the t_0 point their power levels may be 30 to 50 db below the maximum echo power which would be expected if the effects of D and r_0 were negligible. These attenuations may seem large, but they can be overcome by quite practical and not prohibitively expensive increases in transmitter power and antenna gain.

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D.W. R. McKinley

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Preface

Most of us, outdoors on a warm summer night, have taken passing notice of the shooting stars which streak across the sky at random intervals. Perhaps once a year or so, in a given locality, a brilliant fireball may light up the countryside and thereby unleash a flood of telephone calls to newspaper offices. Rarer still are the spectacular displays of celestial fireworks associated with the really great meteor showers, which in the past have aroused emotions ranging from delighted awe to sheer superstitious terror. We know, of course, that these are the visible manifestations of the continual bombardment of our planet by objects from outer space, objects which vary in size from gigantic boulders with the latent energy of an atomic bomb—fortunately these do not occur more frequently than once or twice a century—down to particles the size of pinheads, which create a hundred thousand fleeting sparks a minute all over our globe. Beyond the limits of our telescopically aided vision, we have evidence that large numbers of still finer particles are streaming into the atmosphere; saved from fiery destruction by their very smallness, they drift leisurely down to earth. It may be that, as condensation nuclei, this microscopic meteoric material literally precipitates the gentle rain from heaven.

Although the historical records of meteoric appearances go back for three or four thousand years, it was only around the beginning of the nineteenth century that the first glimmerings of scientific interest showed. Even then, few took up the study with lasting determination, and until comparatively recently, research in meteor science was left to a small but select group of devoted specialists. In the last two decades, though, an increasing awareness of the influence that meteors have on man's activities has blossomed among scientists and engineers and, to a more diffuse extent, among the literate public. More powerful optical tools have been complemented by the new radio observing methods, with the result that rapid progress is being made, both in our fundamental understanding of meteoric phenomena and in our practical utilization of the effects.

In this book we have attempted to summarize the major observational